FARM MANAGEMENT SYSTEM

A Project Report

Submitted by:

Team Leader: KEERTHANA S - (Reg.no:34623U18036)

Team Member: AKSHAYAA P - (Reg.no:34623U18004)

Team Member: CHANDHINI J - (Reg.no:34623U18009)

Team Member: DHIVYA DHARSHINI B - (Reg.no:34623U18014)

Team Member: HARSHITHA R - (Reg.no:34623U18025)

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Hakeem Nagar – Melvisharam

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FARM MANAGEMENT SYSTEM

User Story:

In the Farm Management System (FMS), as a farmer, farm manager, or system administrator, I want to use a centralized Salesforce- based platform to record and manage farm, crop, farmer, and buyer information so that all agricultural data is stored securely and systematically. I want to enter and update farm details, crop production records, and buyer transactions easily through a user friendly interface with proper validation rules to prevent incorrect entries. I also want the system to generate real- time reports and interactive dashboards so that I can analyze farm productivity, monitor sales, and make informed decisions. Additionally, I want role- based access and secure data storage to ensure that sensitive information is protected while enabling smooth collaboration among users. This system will help reduce manual paperwork, minimize errors, improve efficiency, and support data- driven farm management.

1. INTRODUCTION

1.1 Project Overview:

The Farm Management System (FMS) is a Salesforce- based application designed to digitize and streamline agricultural operations. It manages data related to farms, crops, farmers, and buyers while providing real- time insights through reports and dashboards. The system helps farmers and managers monitor agricultural activities efficiently and make informed decisions.

1.2 Purpose :

The main purpose of this project is to:

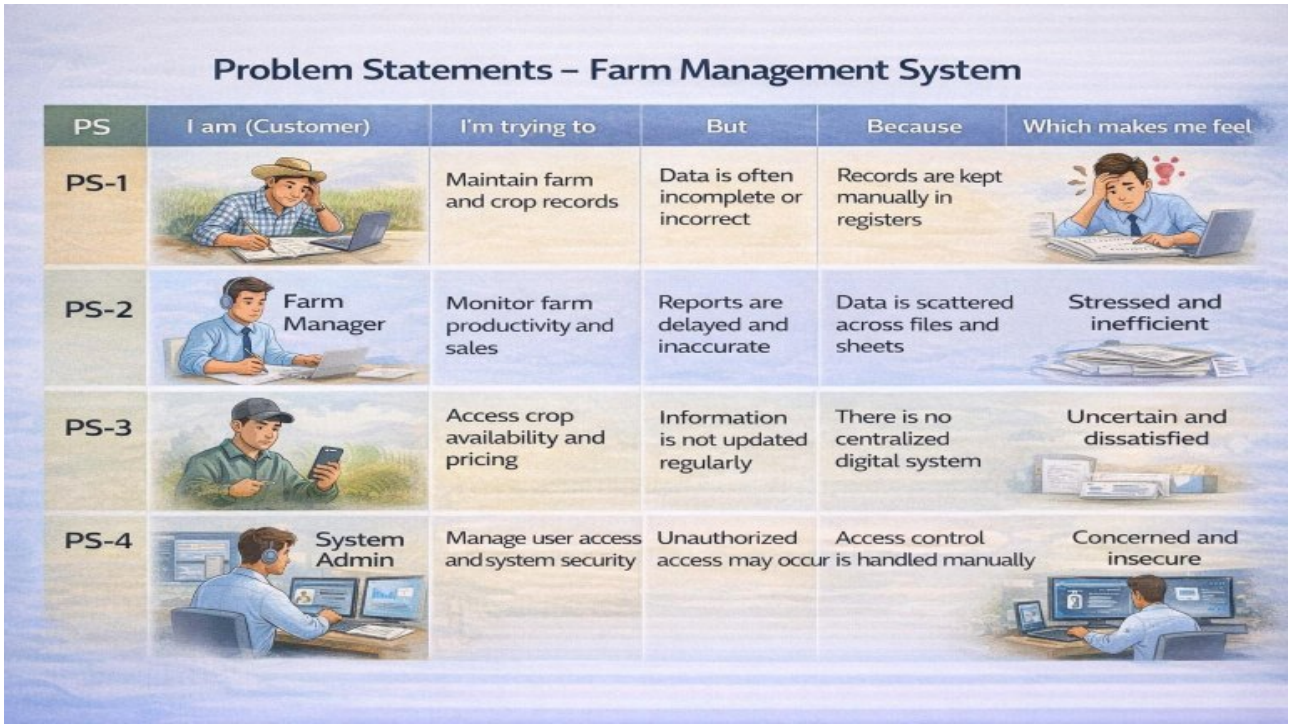
- Digitize agricultural data management
- Improve data accuracy and accessibility
- Automate record handling
- Enable real- time monitoring
- Support data- driven farming decisions
- Enhance productivity and transparency

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel

PS- 1	Farmer	Maintain farm and crop records	Data is often incomplete or incorrect	Records are kept manually in registers	Frustrated and confused
PS- 2	Farm Manager	Monitor farm productivity and sales	Reports are delayed and inaccurate	Data is scattered across files and sheets	Stressed and inefficient
PS- 3	Buyer	Access crop availability and pricing	Information is not updated regularly	There is no centralized digital system	Uncertain and dissatisfied
PS- 4	System Admin	Manage user access and system security	Unauthorized access may occur	Access control is handled manually	Concerned and insecure



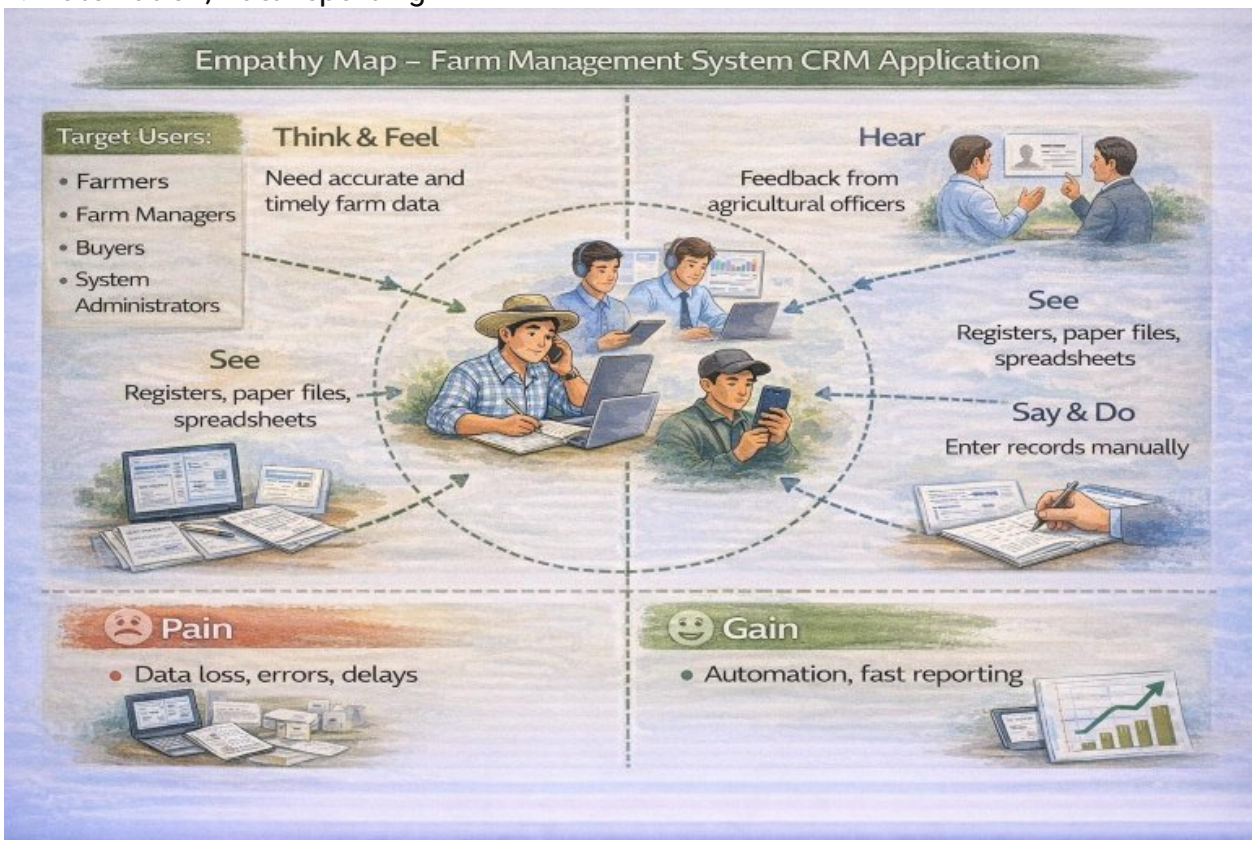
2.2 Empathy Map Canvas

Target Users:

- Farmers
- Farm Managers
- Buyers
- System Administrators

Empathy Analysis

- Think & Feel: Need accurate and timely farm data
- Hear: Feedback from agricultural officers
- See: Registers, paper files, spreadsheets
- Say & Do: Enter records manually
- Pain: Data loss, errors, delays
- Gain: Automation, fast reporting



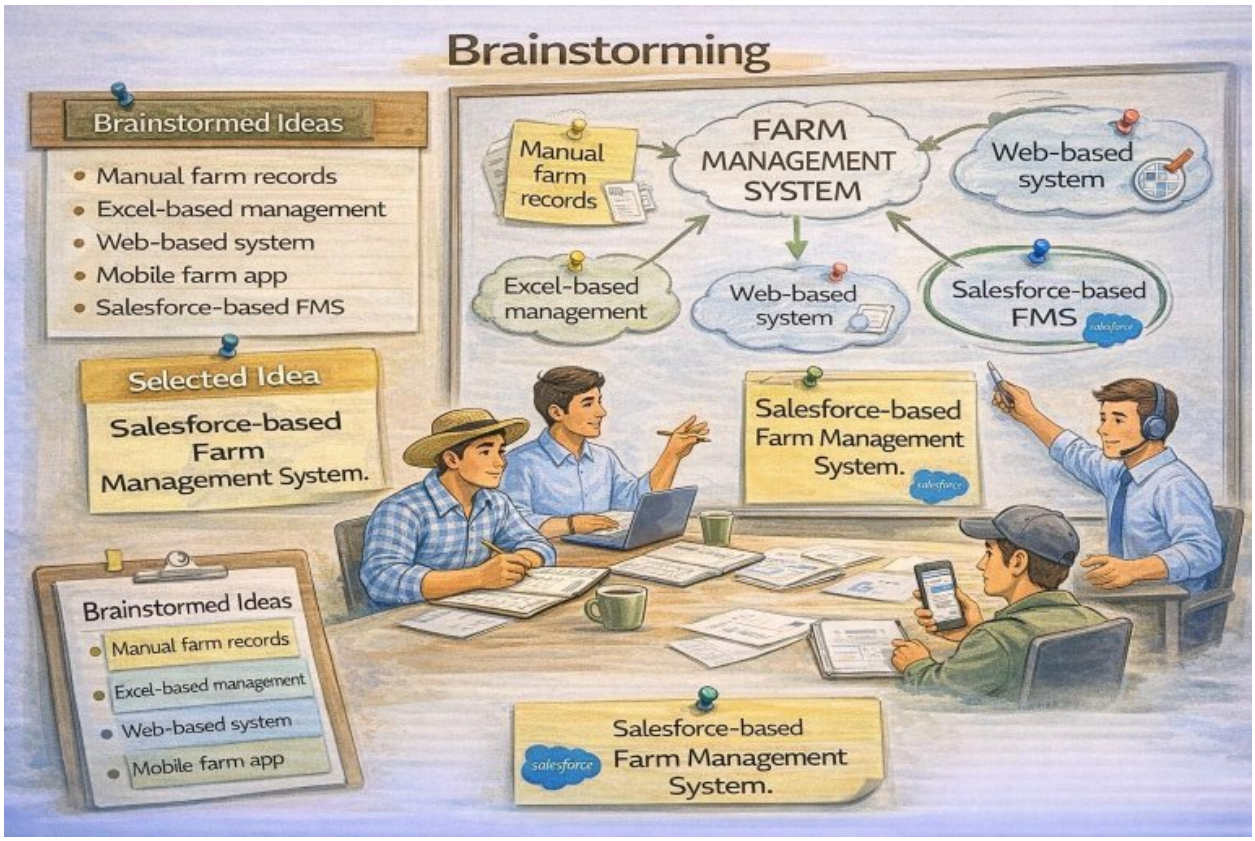
2.3 Brainstorming

Brainstormed Ideas

- Manual farm records
- Excel- based management
- Web- based system
- Mobile farm app
- Salesforce- based FMS

Selected Idea

Salesforce- based Farm Management System.

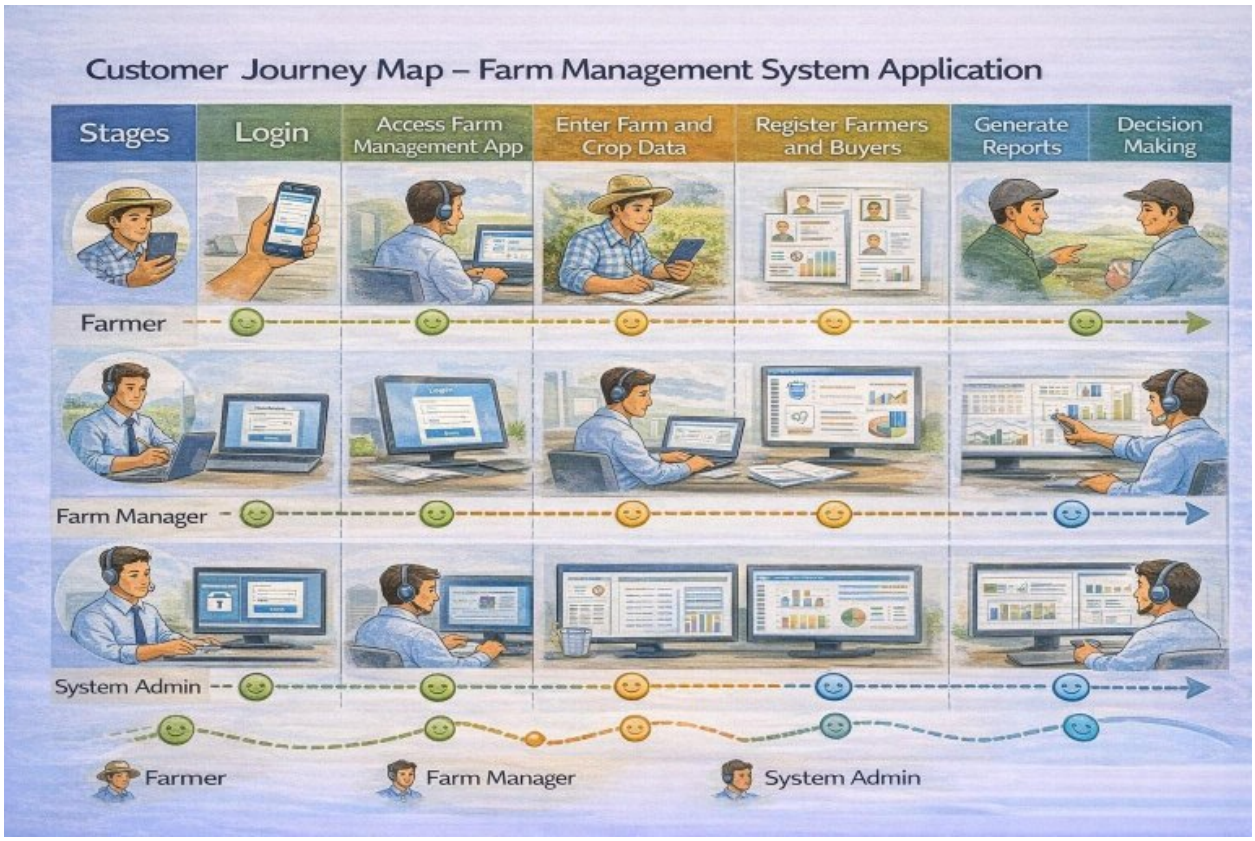


3. REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Stages:

1. Login
2. Access Farm Management App
3. Enter Farm and Crop Data
4. Register Farmers and Buyers
5. Generate Reports
6. Analyze Results 7. Decision Making



3.2 Solution Requirements

Functional Requirements

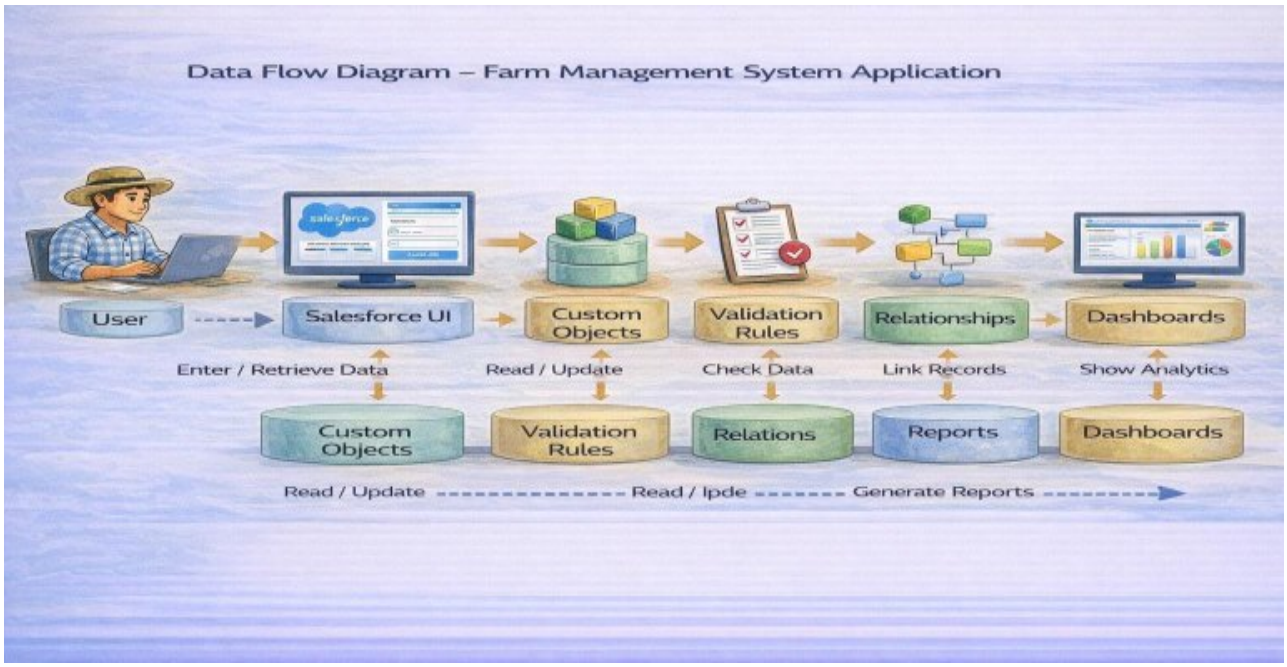
FR No	Requirement
FR- 1	User Authentication
FR- 2	Farm, Crop, Farmer, Buyer Management
FR- 3	Data Entry and Validation
FR- 4	Relationship Mapping
FR- 5	Report Generation
FR- 6	Dashboard Visualization
FR- 7	User Role Management

Non- Functional Requirements

NFR	Description	Example
Usability	Easy interface	Simple Lightning UI
Security	Data protection	Profiles & roles
Reliability	No data loss	Cloud backup
Performance	Fast access	Reports in <3 sec
Availability	24/7 access	Cloud hosting
Scalability	Handles growth	More farms/users

3.3 Data Flow Diagram

Data Flow:
User Salesforce UI Custom Objects (Farm, Crop, Farmer, Buyer) Validation Rules Relationships Reports Dashboards



3.4 Technology Stack

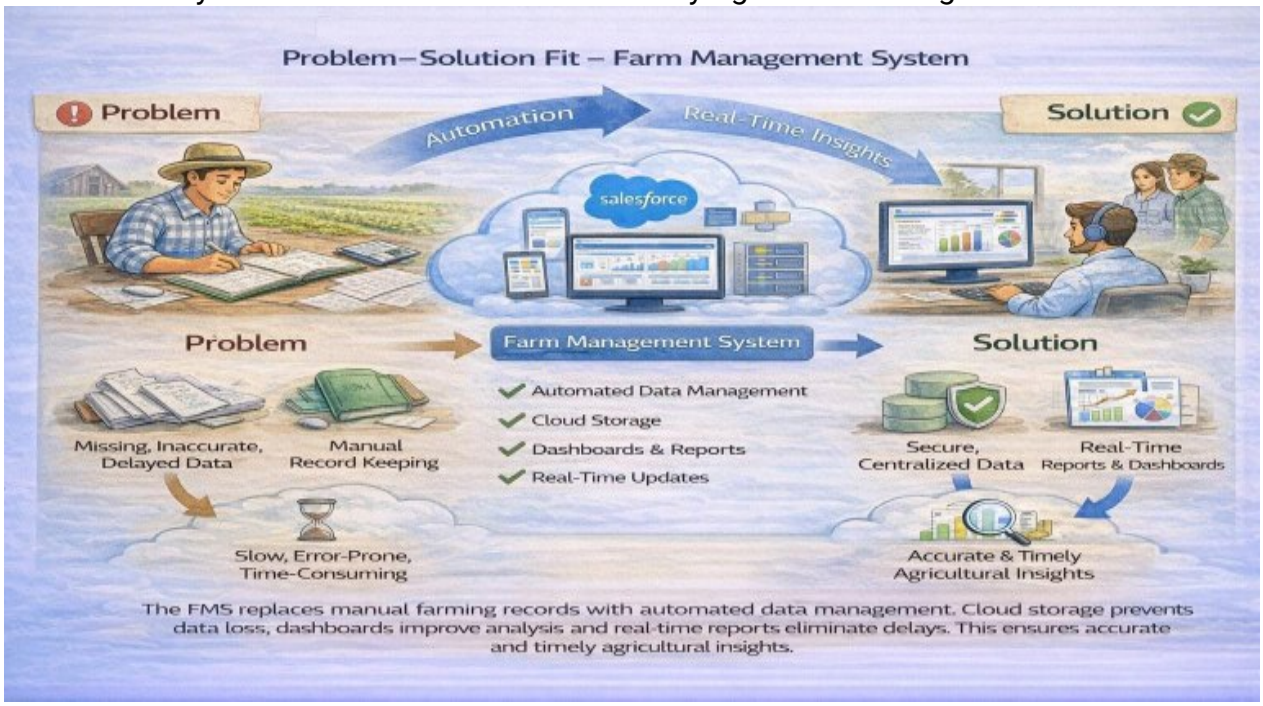
Layer	Technology

Frontend	Salesforce Lightning UI
Backend	Salesforce Platform
Database	Salesforce Cloud
Automation	Validation Rules
Analytics	Reports & Dashboards

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

The FMS replaces manual farming records with automated data management. Cloud storage prevents data loss, dashboards improve analysis, and real- time reports eliminate delays. This ensures accurate and timely agricultural insights.



4.2 Proposed Solution

Functional Requirements

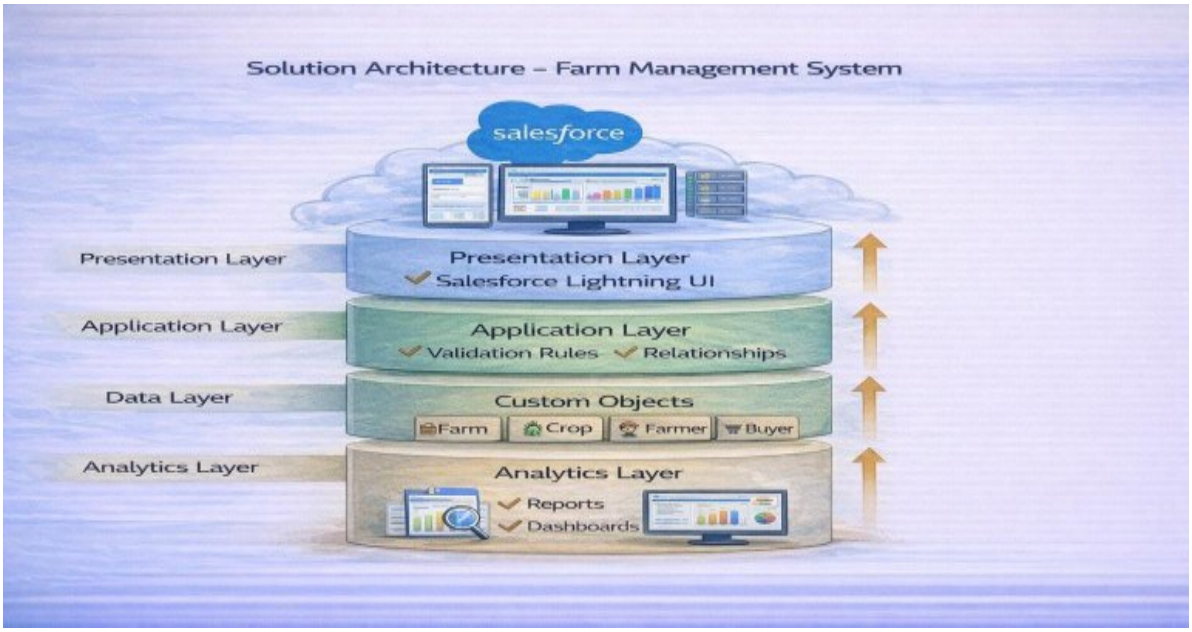
Parameter	Description

Problem	Manual maintenance of farm, crop, farmer, and buyer records leading to errors, data loss, and delays.
Solution	Salesforce- based Farm Management System with centralized data storage and automation.
Innovation	Automated data entry validation, cloud storage, and real-time reporting dashboards.
Impact	Improved data accuracy, faster reporting, and better decision-making for farmers and managers.es
Business Model	Digital farm management and agricultural data services platform.
Scalability	Supports multiple farms, users, and regions with cloud- based infrastructure.

4.3 Solution Architecture

Architecture Layers

1.Presentation Layer – Salesforce Lightning UI
2.Application Layer – Validation Rules, Relationships
3.Data Layer – Custom Objects
4.Analytics Layer – Reports and Dashboards



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- Created Salesforce Developer Org
- Developed custom objects
- Configured tabs and Lightning App
- Created fields and relationships
- Designed profiles and roles
- Created users and sharing rules
- Developed reports and dashboards
- Conducted testing and reviews

Product Backlog, Sprint Schedule, and Estimation (4 Marks)
Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Developer Setup	USN-1	As a system administrator, I want to create and configure a Salesforce developer account so that I can set up and deploy the Farm Management System.	3	High	Member1
Sprint-2	Data Modeling	USN-2	As an admin, I want to create Farm, Crop, Farmer, and Buyer objects with proper relationships so that agricultural data is stored and managed efficiently.	5	High	Member2
Sprint-2	Data Modeling	USN-3	As a staff member, I want tabs and a Lightning App for all farm modules so that I can easily navigate and perform daily operations.	5	High	Member3
Sprint-3	Automation	USN-4	As a system, I want to validate farm and crop details such as dates, quantities, and mandatory fields so that incorrect or incomplete records are prevented.	3	High	Member4

Sprint- 3	Automation	USN- 5	As a manager, I want automated flows for crop updates and buyer notifications so that farm operations are processed faster without manual intervention.	3	High	Member 4
Sprint- 4	Security	USN- 6	As an administrator, I want role- based access control for farmers, managers, and buyers so that sensitive farm and transaction data is secure.	5	High	Member 2
Sprint- 5	Reports	USN- 7	As a farm manager, I want reports showing crop yield, sales, and buyer transactions so that I can monitor farm performance and make informed decisions.	4	High	Member 3
Sprint- 6	Dashboards	USN- 8	As a farm manager, I want dashboards showing crop productivity, sales trends, and operational statistics so that I can monitor performance visually.	4	Medium	Member 4

6. Project Development Phase:

Project Description:

The Farm Management System (FMS) developed using Salesforce is designed to help farmers and farm managers manage agricultural activities in a digital and organized way. This system stores and manages

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	24 Oct 2022	29 Oct 2022	20	29 Oct 2022
Sprint-2	20	6 Days	31 Oct 2022	05 Nov 2022		
Sprint-3	20	6 Days	07 Nov 2022	12 Nov 2022		

Sprint-4	20	6 Days	14 Nov 2022	19 Nov 2022		

important information related to farms, farmers, crops, and buyers in one centralized platform.

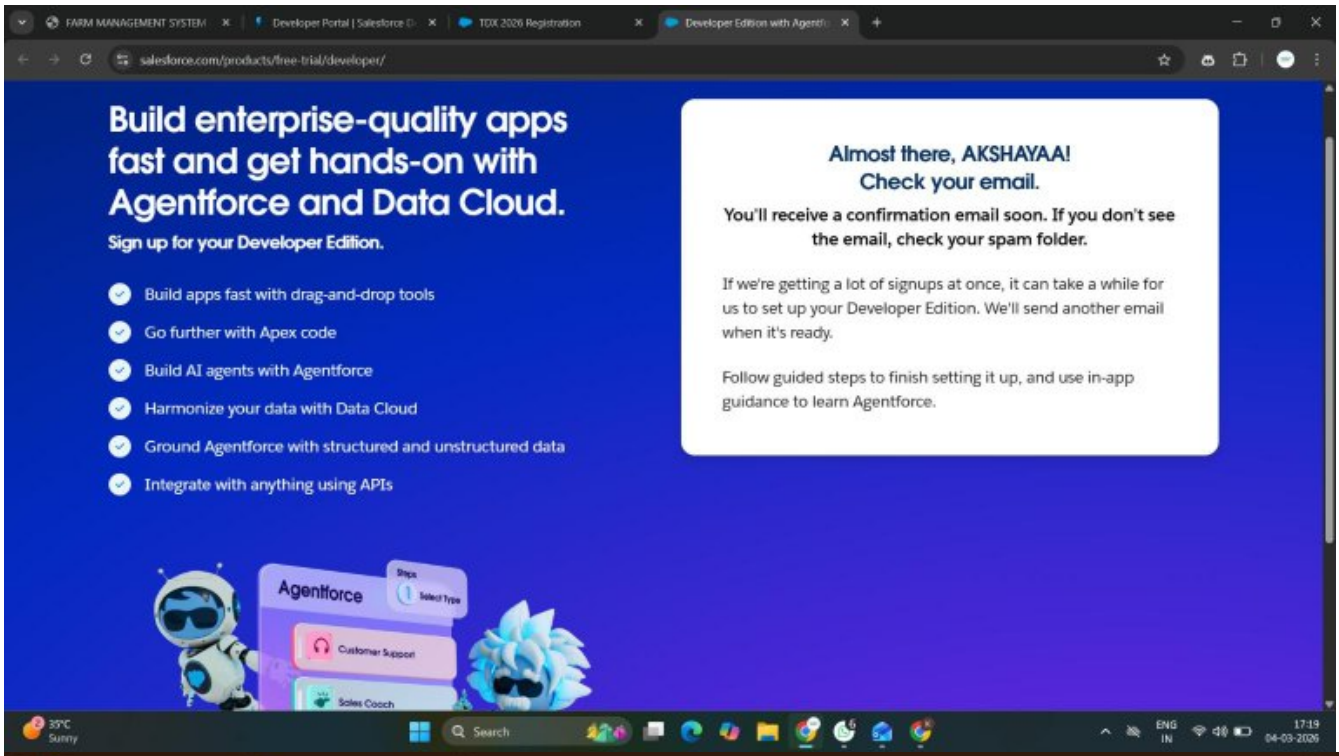
The application helps users keep records of different farms and the crops grown in each farm. It also maintains the details of farmers who manage the farms and the buyers who purchase agricultural products. By storing all this information in the system, users can easily track farming activities and maintain proper data management. The Farm Management System also uses Salesforce features like reports and dashboards to provide useful insights about crop production, farm performance, and buyer information. These insights help farm managers make better decisions related to crop planning, sales, and resource management.

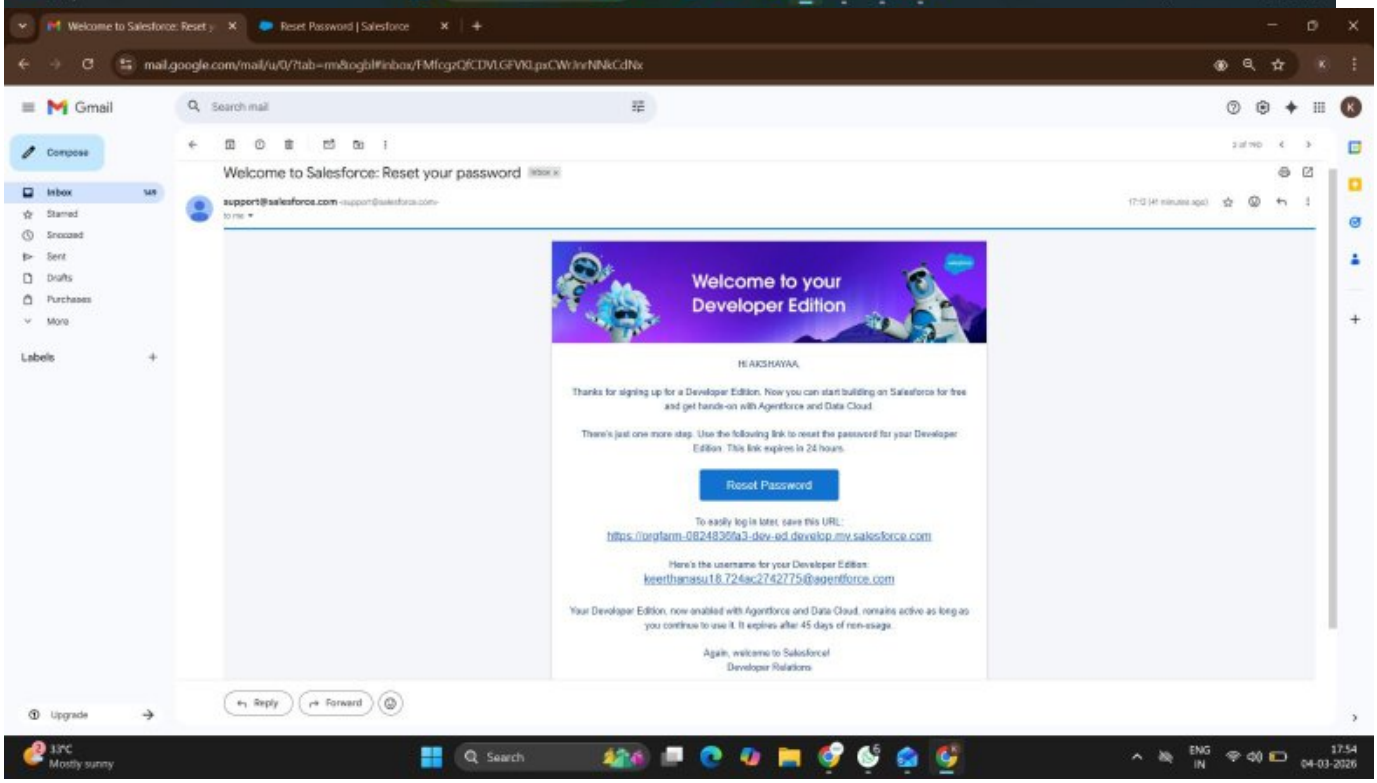
Milestone 1- Salesforce:

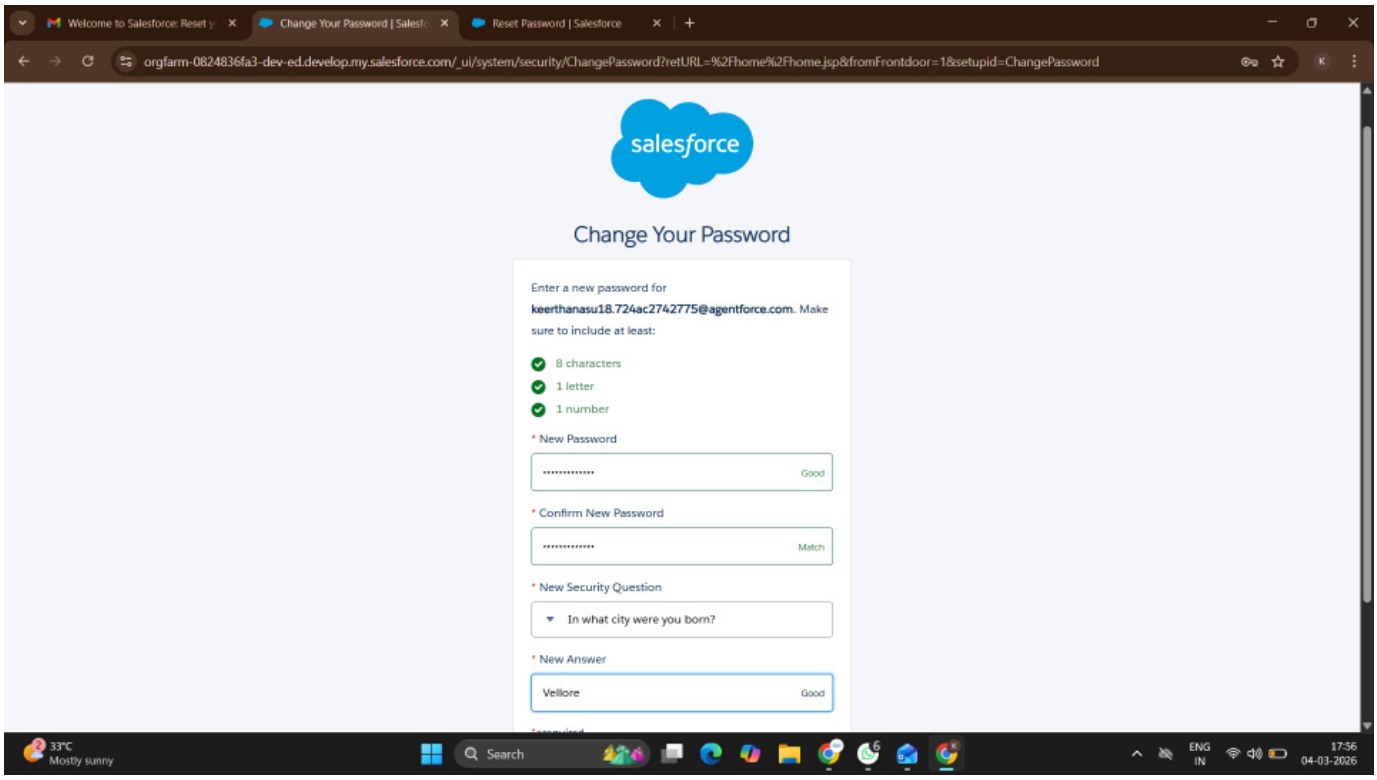
What Is Salesforce?
Salesforce is your customer success platform, designed to help you sell, service, market, analyze, and connect with your customers.
Salesforce has everything you need to run your business from anywhere. Using standard products and features, you can manage relationships with prospects and customers, collaborate and engage with employees and partners, and store your data securely in the cloud.

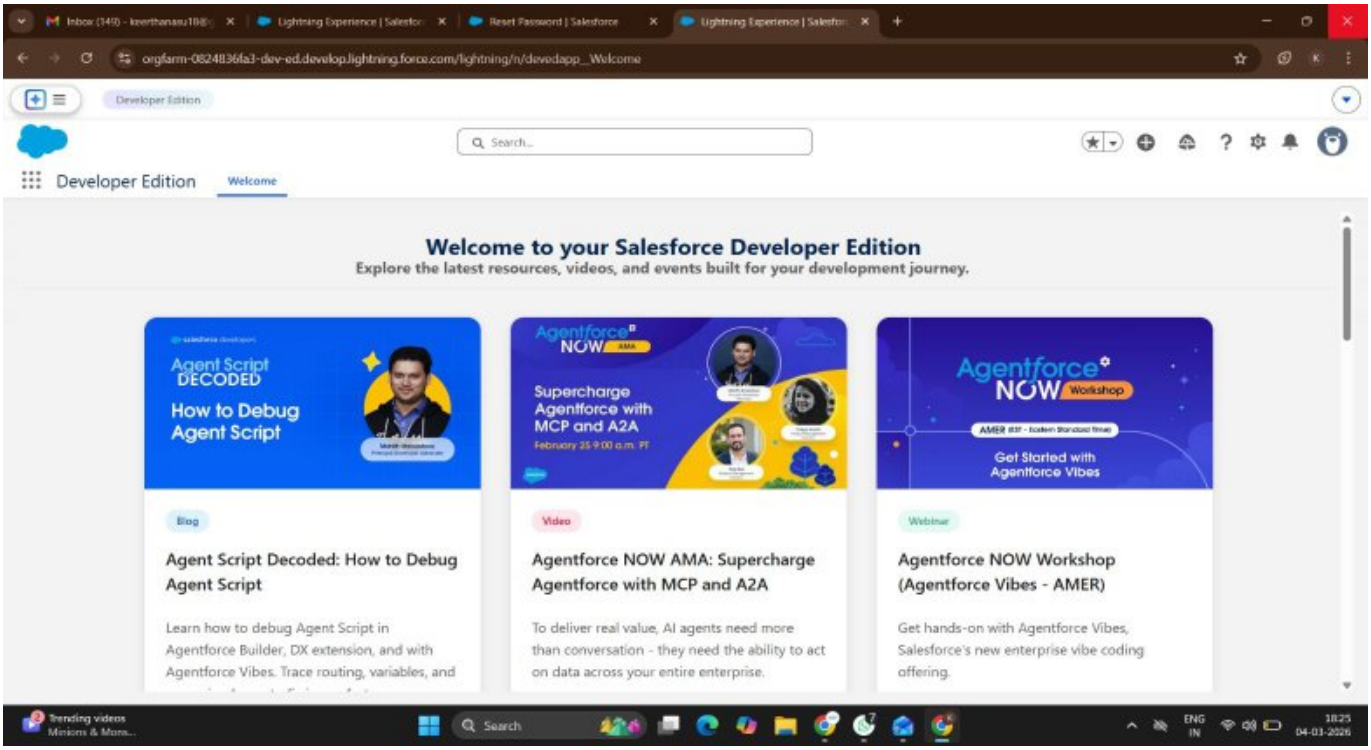
Activity1:
Creating a Salesforce Developer Org:
A Developer org has all the features and licenses you need to get started with Salesforce.

1. Search Developer.salesforce.com/signup
2. Enter the following details like First name, last name, Email, Role, Company, Country/Region, Postal code, and Username must be unique.
3. Click sign me up, after a few min you will reserve a mail salesforce org and by using the verify account link you can create your new password.
4. Click save.
5. Search login.salesforce.com
6. By using username and password you can into the Salesforce org. The homepage will appear below


salesforce.com/products/free-trial/developer/
Build enterprise-quality apps fast and get hands-on with Agentforce and Data Cloud.
Sign up for your Developer Edition.
Build apps fast with drag-and-drop tools
Go further with Apex code
Build AI agents with Agentforce
Harmonize your data with Data Cloud
Ground Agentforce with structured and unstructured data
Integrate with anything using APIs
Almost there, AKSHAYAA!
Check your email.
You'll receive a confirmation email soon. If you don't see the email, check your spam folder.
If we're getting a lot of signups at once, it can take a while for us to set up your Developer Edition. We'll send another email when it's ready.
Follow guided steps to finish setting it up, and use in-app guidance to learn Agentforce.
Agentforce
Customer Support
Sales Coach


Welcome to Salesforce: Reset your password
support@salesforce.com
Welcome to your Developer Edition
Hi AKSHAYAA,
Thanks for signing up for a Developer Edition. Now you can start building on Salesforce for free and get hands-on with Agentforce and Data Cloud.
There's just one more step. Use the following link to reset the password for your Developer Edition. This link expires in 24 hours.
Reset Password
To easily log in later, save this URL:
https://orgfarm-0824836fa3-dev-ed.develop.my.salesforce.com
Here's the username for your Developer Edition:
keerthanasu18.724ac2742775@agentforce.com
Your Developer Edition, now enabled with Agentforce and Data Cloud, remains active as long as you continue to use it. It expires after 45 days of non-usage.
Again, welcome to Salesforce!
Developer Relations
Reply
Forward





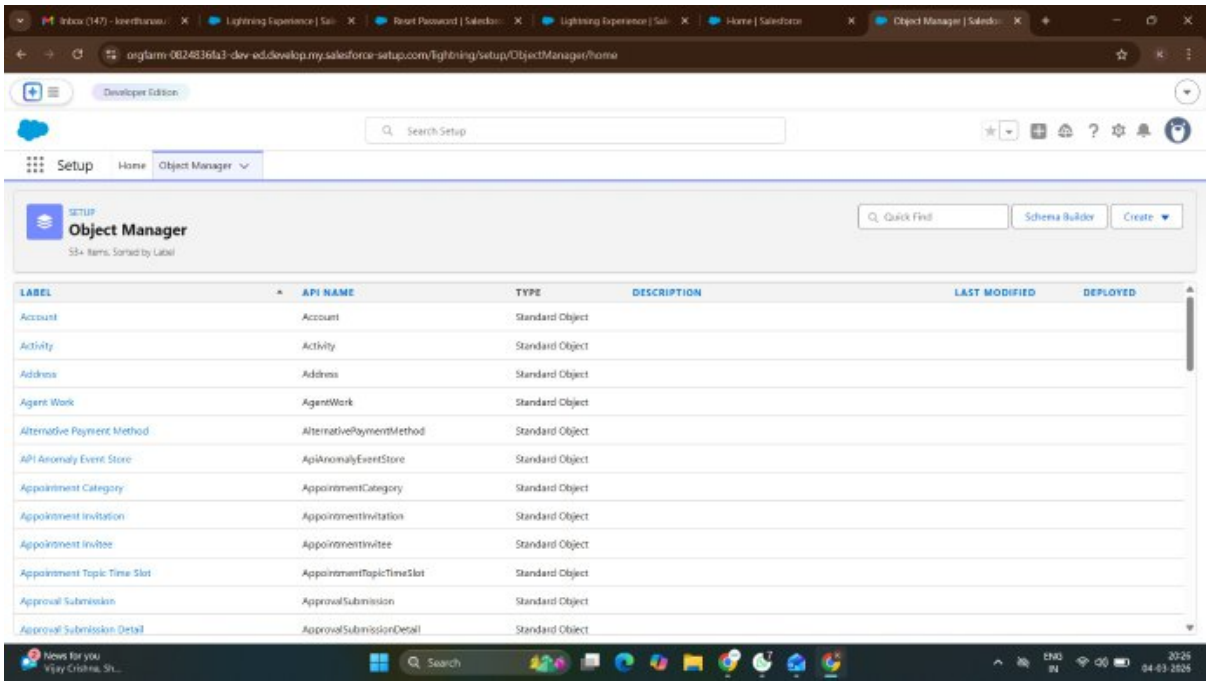
Milestone 2- Object:

What is an object?
Salesforce objects are database tables that permit you to store data that is specific to an organization. It consists of fields (columns) and records (rows).

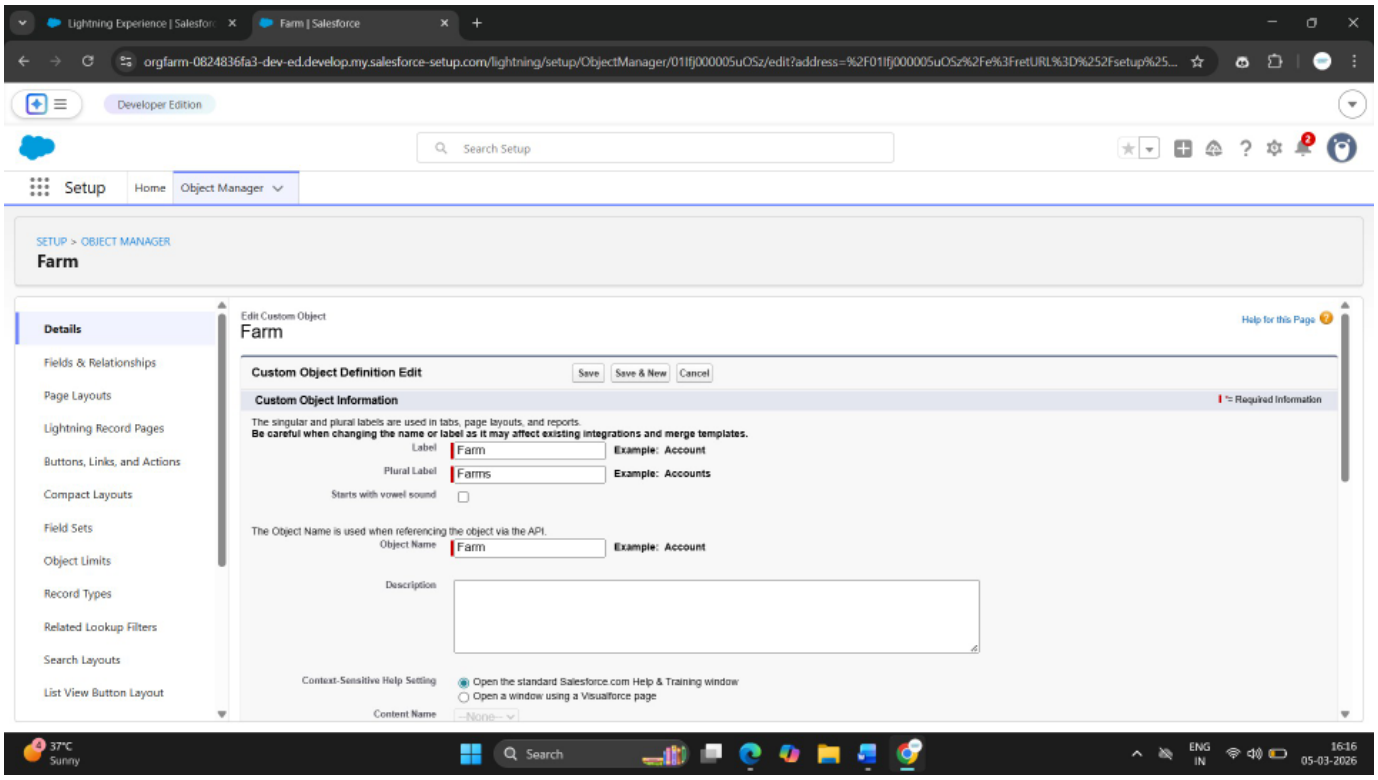
In This Application We Use 4 Custom Objects:
1. Farm
2. Crop
3. Farmer
4. Buyer

Activity 1:

Create a custom object for Farm:
To create a custom object, follow these steps:
1. From setup click on object manager.
2. Click create, select custom object.



3. Fill in the label as " Farm ".
4. Fill in the plural label as " Farms".
5. Record name: Farm Name"
6. Select the data type as " Text" .



7. In the Optional Features section, select Allow Reports and Track Field History.
8. In the Deployment Status section, ensure Deployed is selected.
9. In the Search Status section, select Allow Search.
10. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.
11. Leave everything else as is, and click Save.

Activity 2:

Creation of Crop object:

1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.

a. On the Custom Object Definition page, create the object as follows: b. Label: Crop

c. Plural Label: Crops

d. Record Name: Crop Name

e. Select the data type as " Text" .

f. Check the Allow Reports checkbox g. Check the Allow Search checkbox

h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout i. Click Save.

Activity 3:

Creation of Farmer object:

1.Click on the gear icon and then select Setup.
2.Click on the object manager tab just beside the home tab.
3.After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.
a.On the Custom Object Definition page, create the object as follows:
b.Label: Farmer
c.Plural Label: Farmers
d.Record Name: Farmer Name
e.Select the data type as "Text".
f.Check the Allow Reports checkbox.
g.Check the Allow Search checkbox.
h.In the Object Creation Options section, select Add Notes and Attachments related list to default page layout. i. Click Save.

Activity 4:

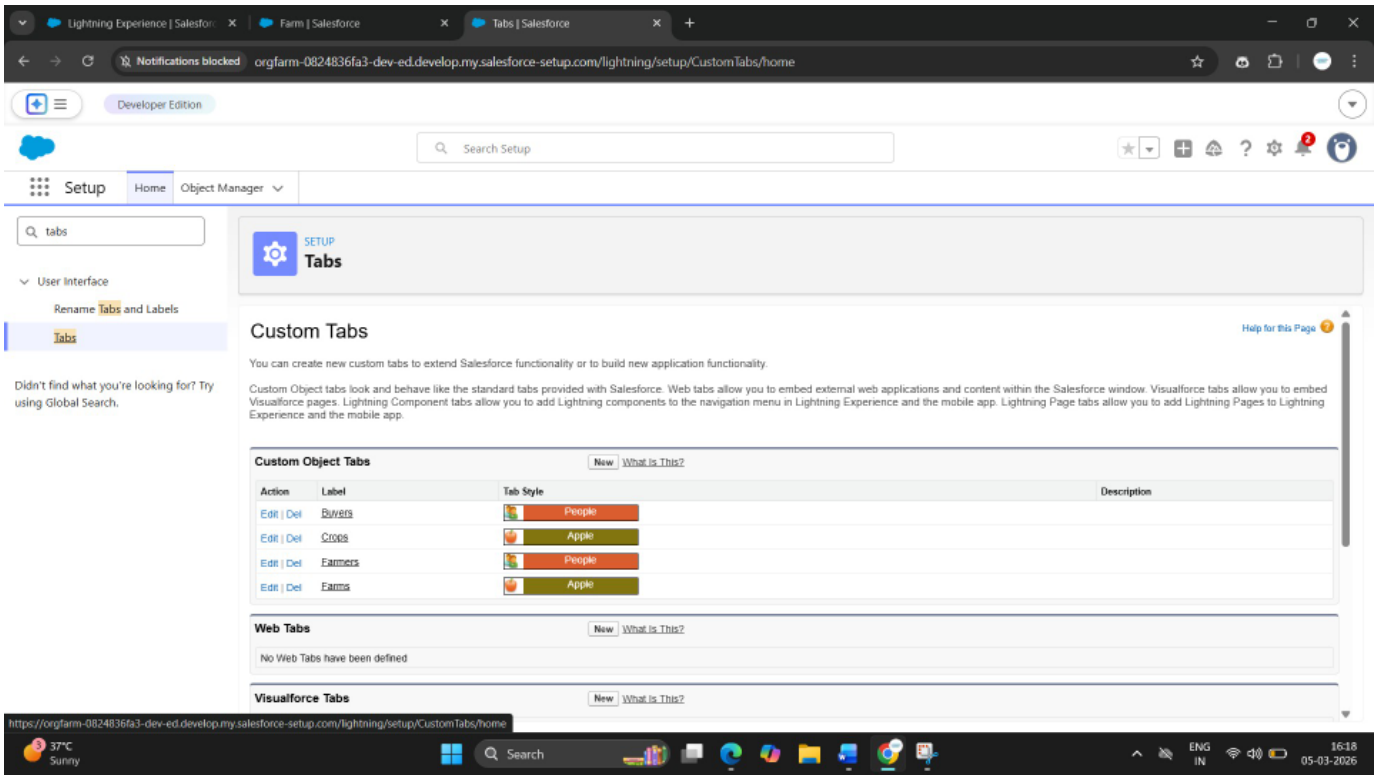
Creation of Buyer object:
1.Click on the gear icon and then select Setup.
2.Click on the object manager tab just beside the home tab.
3.After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.
a. On the Custom Object Definition page, create the object as follows: b. Label: Buyer
c.Plural Label: Buyers
d.Record Name: Buyer Name
e.Select the data type as "Text".
f.Check the Allow Reports checkbox.
g. Check the Allow Search checkbox.
h. In the Object Creation Options section, select Add Notes and Attachments related list to
default page layout.
i. Click Save.

Milestone 3 - Tab:

Tab
Tabs in Salesforce help users view the information at a glance. It displays the data of objects and other web content in the application.

Activity 1:

Creation of Farm tab:
Now create a custom tab.
1.Click on the Home tab, enter Tabs in Quick Find and select Tabs.



2.Under custom object tabs, click New.
3.For Object, select Farm.
4.For Tab Style, select any icon.
5.Leave all defaults as is. Click Next, Next, and Save.

Activity 2:

Creation of Crop tab:
Now create a custom tab.
1.Click on Home tab, enter Tabs in Quick Find and select Tabs.
2.Under custom object tabs, click New.
3.For Object, select Crop.
4.For Tab Style, select any icon.
5.Leave all defaults as is. Click Next, Next, and Save.

Activity 3:

Creation of Farmer tab:
Now create a custom tab.
1.Click on the Home tab, enter Tabs in Quick Find and select Tabs.
2.Under custom object tabs, click New.
3.For Object, select Farmer.
4.For Tab Style, select any icon.
5.Leave all defaults as is. Click Next, Next, and Save.

Activity 4:

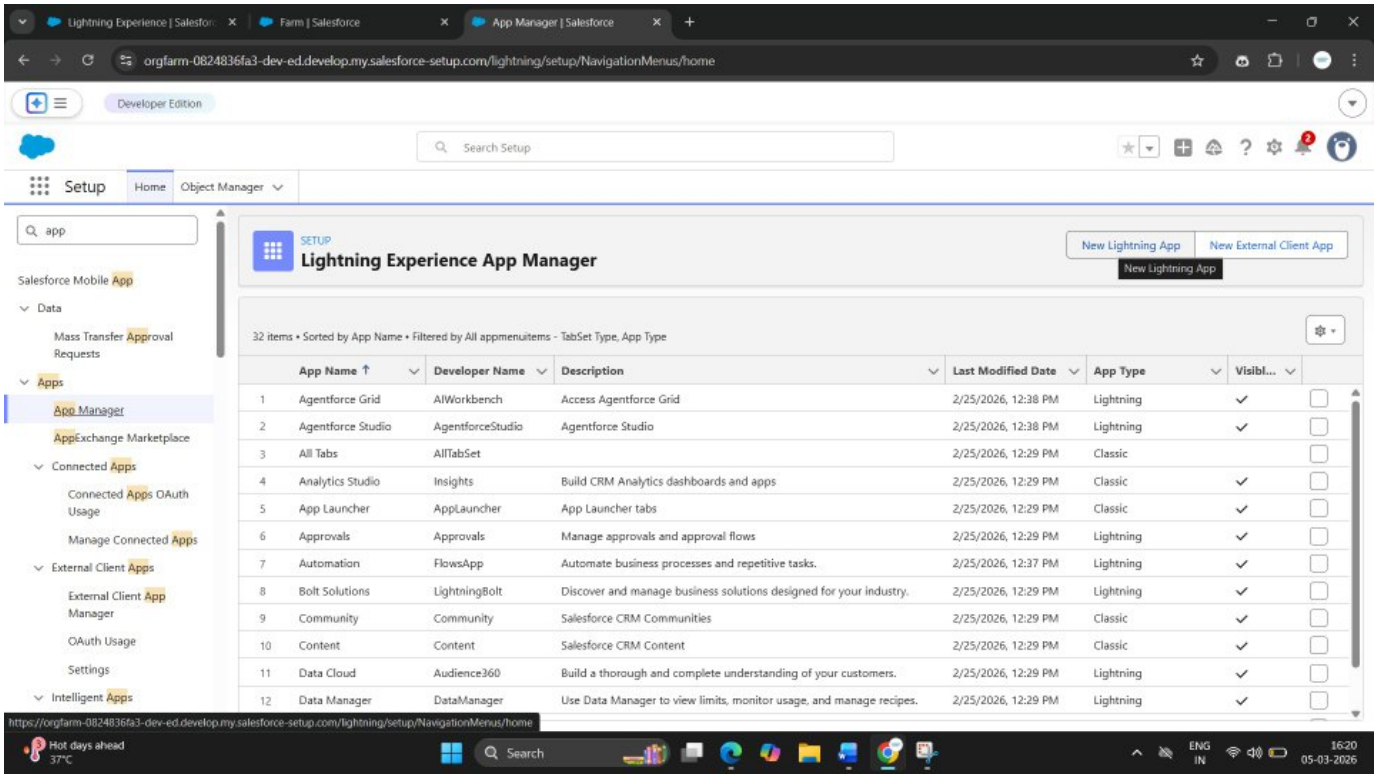
Creation of Buyer tab:
Now create a custom tab.

1. Click on Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Buyer
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Milestone 4 - Lightning App:

What is an App?
Apps in Salesforce are a group of tabs that help the application function by working together as a unit. It has a name, a logo, and a particular set of tabs. The simplest app usually has just two tabs.

Activity 1:
Create the Farm Management System app

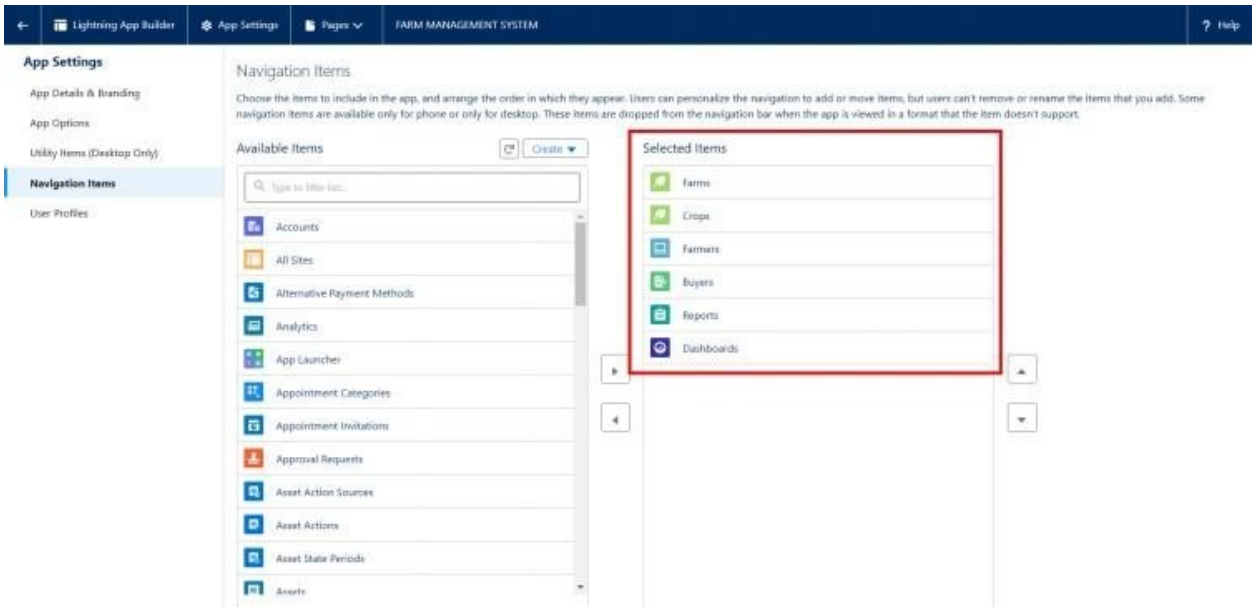


- From Setup, enter App Manager in the Quick Find and select App Manager.
- Click New Lightning App. Farm Management System as the App Name, then click Next Under App Options, leave the default selections and click Next.
- Under Utility Items, leave as is and click Next.
- From Available Items, select Farm, Crop, Farmer, Buyer Reports, and Dashboards and move them to Selected Items. Click Next. From Available Profiles, select System Administrator and move it to Selected Profiles. Click Save & Finish.

- To verify your changes, click the App Launcher, type Farm Management System and select the Job Application app.

Note:

1. App Launcher- Displays available apps.
2. App Name- Displays the current selected app.
3. Navigation menu- Displays the tabs available inside the app.



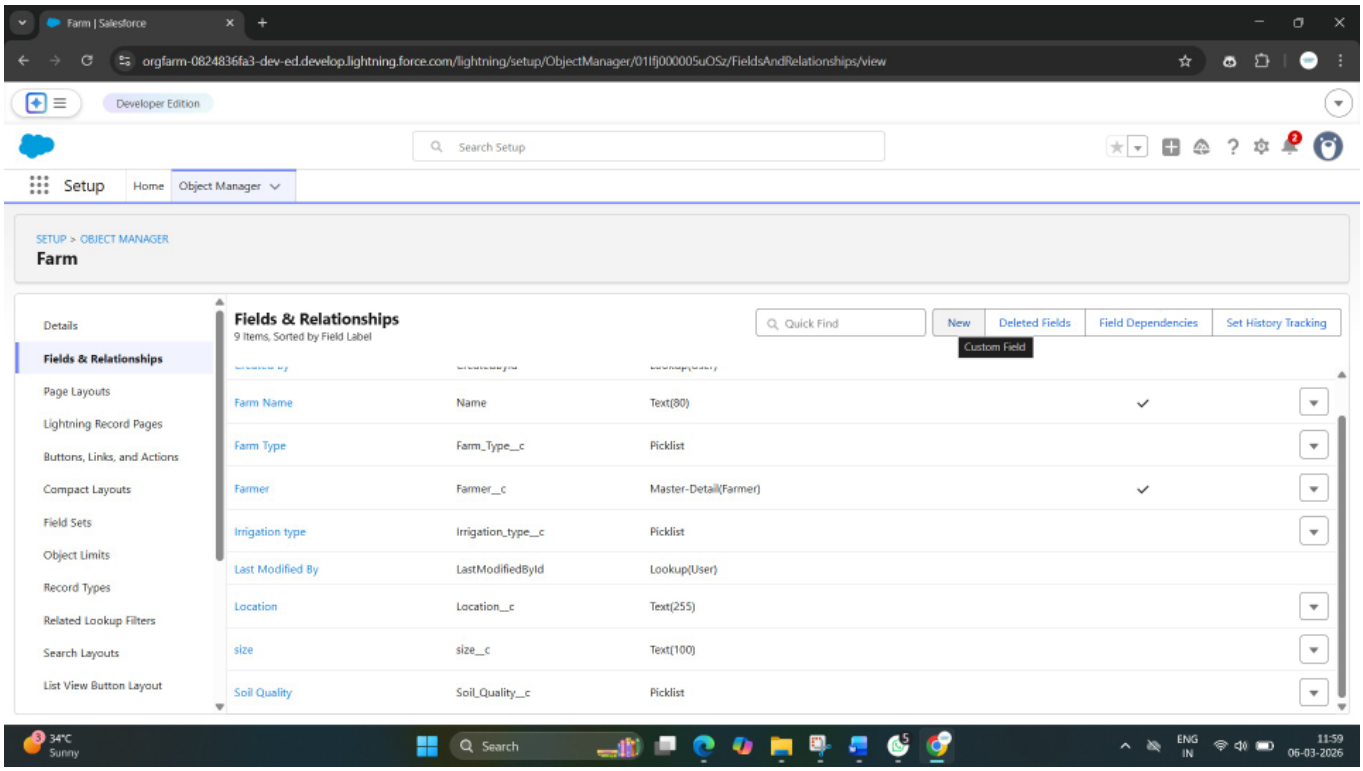
Milestone 5- Fields and Relationship:

What are fields?
Fields in Salesforce represent what the columns represent in relational databases. It can store data values which are required for a particular object in a record.

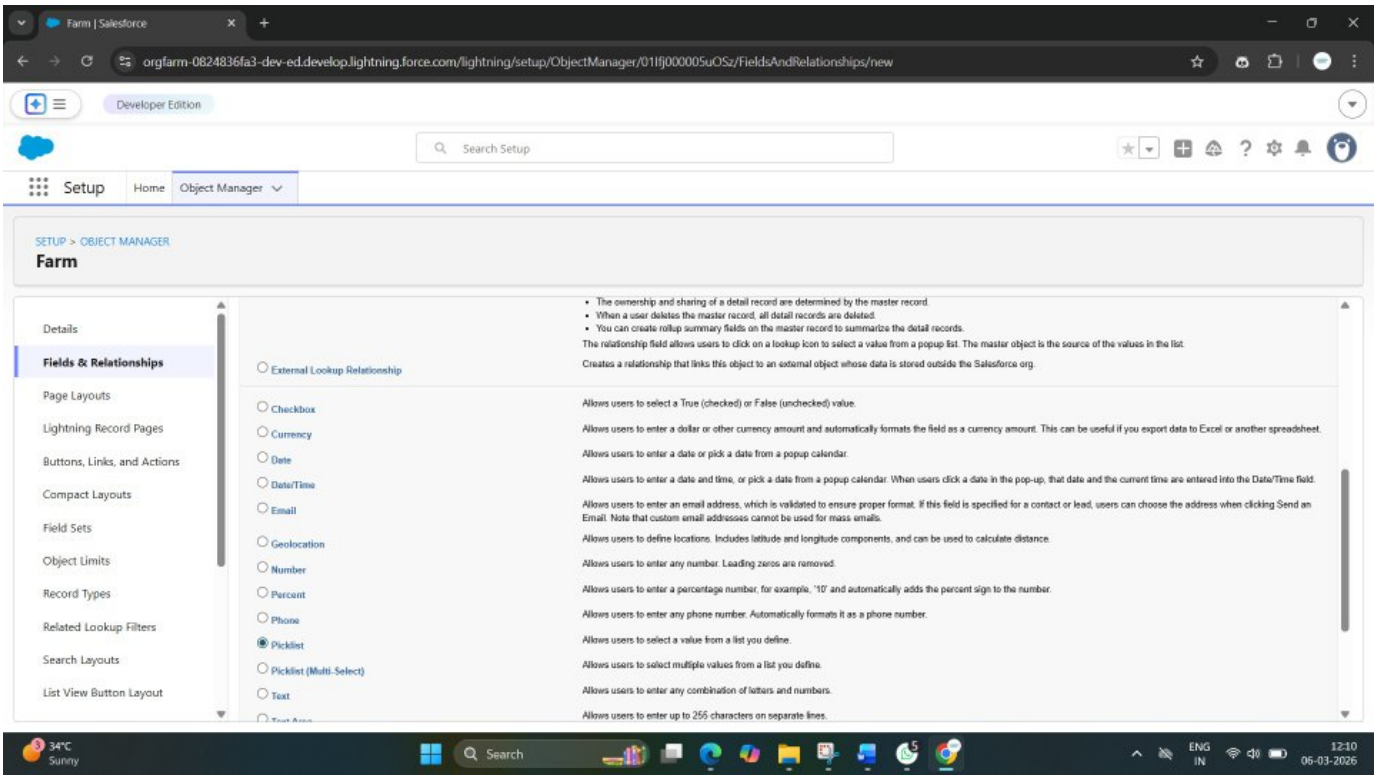
Activity 1:

Creation of fields for the Farm objects:

1. click the gear icon and select Setup. This launches Setup in a new tab.
2. Click the Object Manager tab next to Home.
3. Select Farm.
4. Select Fields & Relationships from the left navigation, and click New



5.Choose Data type as a picklist, Click next



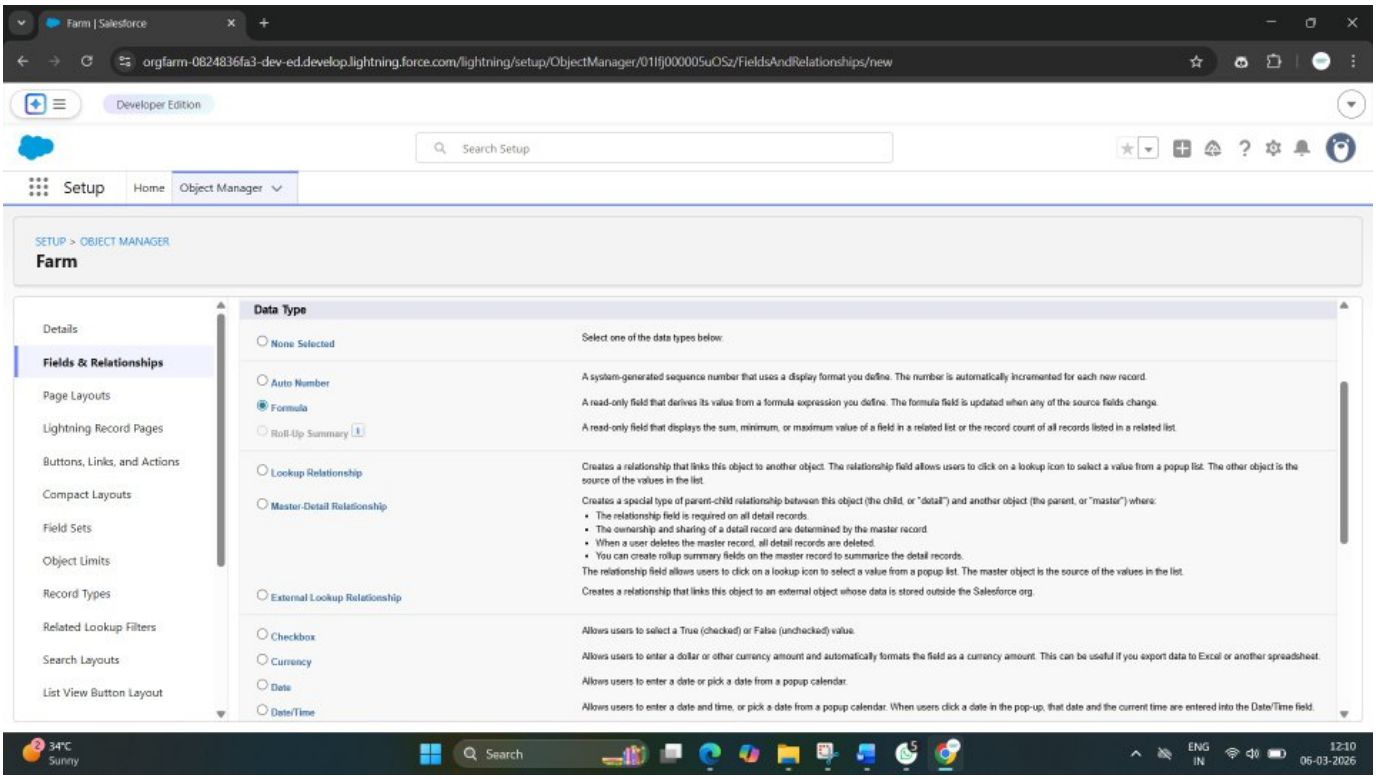
6.Enter Field Label (Farm Type), and Enter Values as (Crop, Livestock, Mixed)
7.Next, Next Click Save.
Now let's create the other fields following Activity1 steps 1 to 4 and we must choose the data types

of the fields carefully.

1. Next we Create another Picklist Field, Enter field label (Soil Quality) enter values (Organic, Cover Crops, Composting)
2. Again we Create Picklist field, Field Label (Irrigation type), Enter Values (Sprinkler irrigation, Drip irrigation, localized irrigation)
3. Choose data type as a text, enter field label (Location)
4. Choose data type as text, enter field label (size)
5. Create a Master Detail relationShip with Farmer Object.

Activity 2:

Creation of fields for the Crop objects:
1. Choose Data type As a text, Enter Field Label name (Crop Name), Next save.
2. Select Data type as date, Enter Field label (Planting Date), then save.
3. Again we select data type as a date, Enter field labe (Harvest date), save
4. Next We Create Lookup relationship on crop object realted to Farm Object.
5. We select adata type as currency, Enter field Label (Actual Yield), Next Save it.
6. Select Data type as a Number, enter Field Label (Total Acres Harvested), Enter Maximum Length (20) and save.
7. Now We create Formula Field For That Select data Type As a Formula, Click Next.



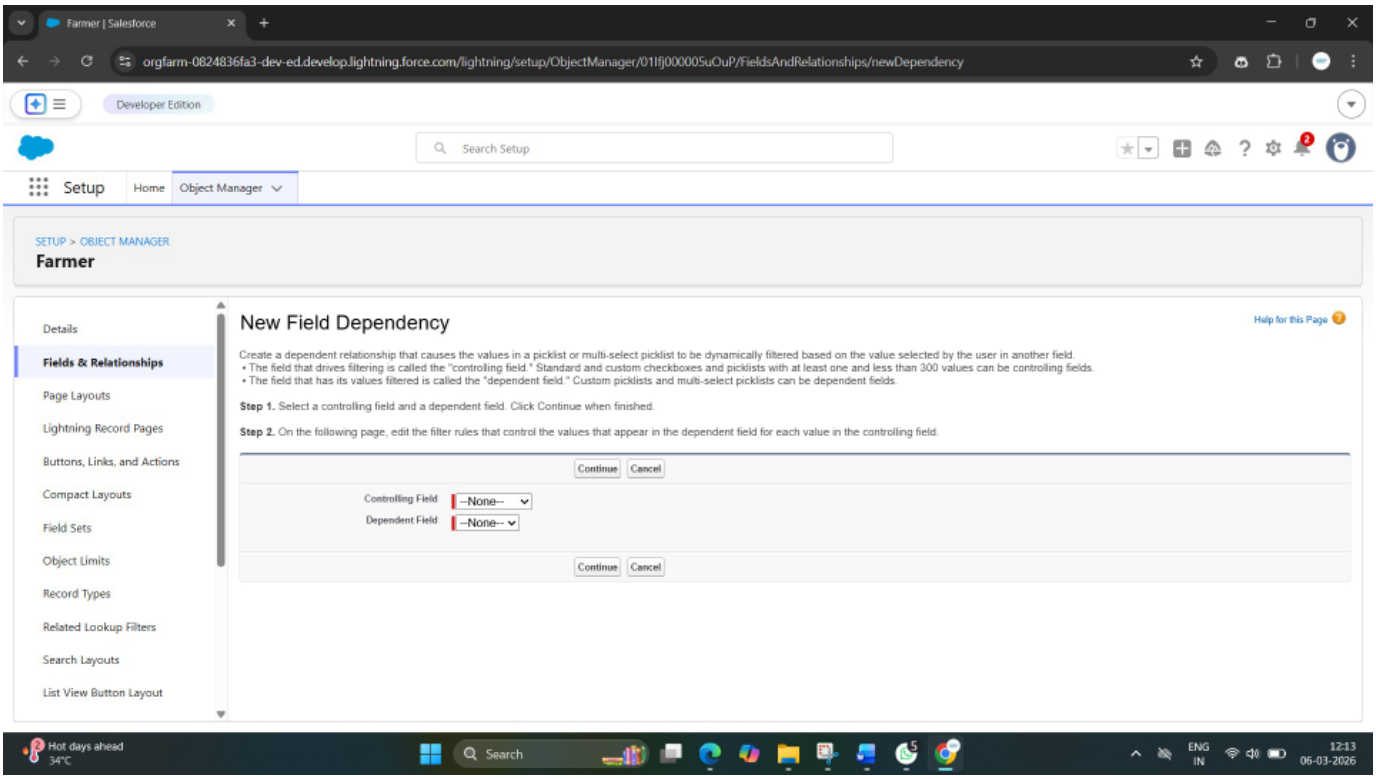
8. Enter Field Label (Expected Yield per Acre), Select formula return type as Currency, click Next.
9. Enter a Formula Actual Yield Divide By Total Acre Harvested, then click next and save.

Activity 3:

Creation of fields for the Farmer objects:
1.Select Data type Phone,Enter field label(Contact Number),next Save.
2.Select Data type Picklist,Enter Field Label(State),Enter values(Telangana,Andhra Pradesh,Tamilnadu),save.
3.Choose Data type Picklist,Enter Field Label (District),Enter Values of district,next save.
4.Again select Data type as Picklist,Enter Field Label (Village),Enter Values of villages,next save.

Creation of field dependency :

1.Now create Field Dependency For District field,Open Object Manager,Choose Farmer object ,select Fields&Relationships,On Right side We have Field Dependency option,select that option.
2.Click new, Select Controlling Field as State,Dependent Field As District,Click on Continue.
3.Select the appropriate Value in each column by double- clicking them.
4.For Ex. Telangana - Yadadri Bhongir.
5.Click Include Values. And it is also the same for Andhra Pradesh, Tamilnadu with its city.
6.Click Preview, then test the dependency by selecting a different State and viewing the associate Values available for Particular state.



7. Click Close to close the preview window, Click Save.

Activity 4:

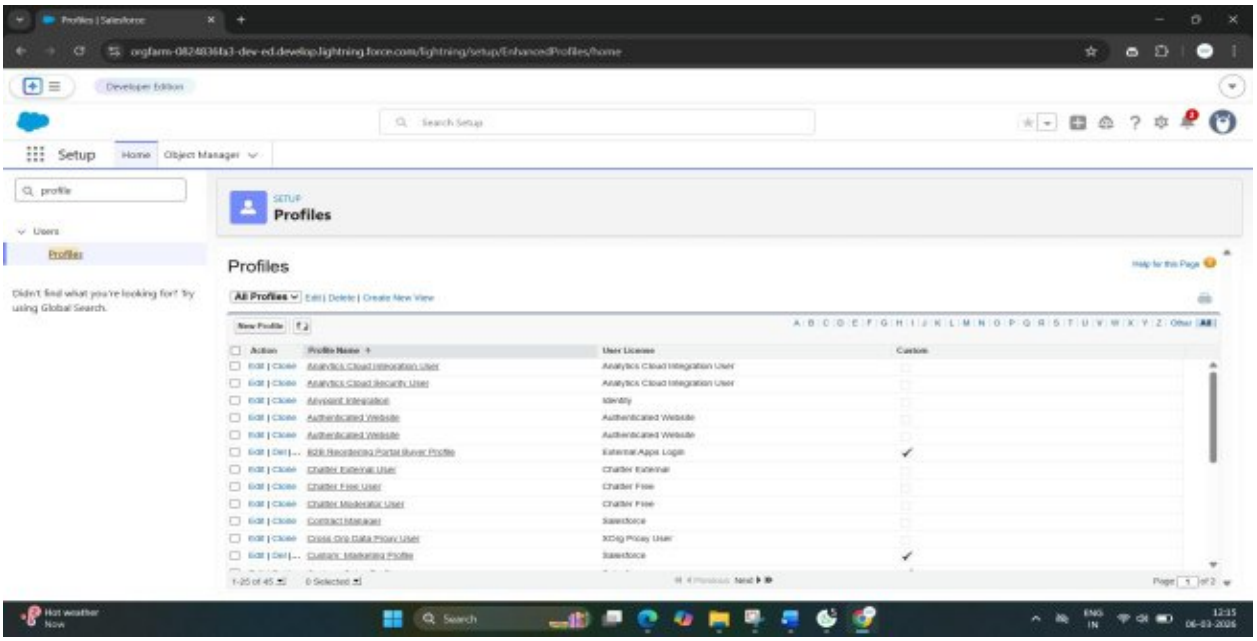
Creation of fields for the Buyer objects:
1.Select Data type as a Phone,Enter a Field Label(Contact),Save.
2.Select Data type as an Email,Enter Field Label(Email), Next save.
3.Select Data type Text area,Enter Field Label(Address),Save it.
4.Select Data type Picklist, Field Label(Buyer Type),Enter values (Wholesale,Retail),Click Save.

Milestone 6- Profile:

What is a profile?
A profile is a group/collection of settings and permissions that define what a user can do in salesforce. A profile controls "Object permissions, Field permissions, User permissions, Tab settings, App settings, Apex class access, Visualforce page access, Page layouts, Record Types, Login hours & Login IP ranges. A profile can be assigned to many users, but user can be assigned single profile at a time.

Activity 1:

Create a custom profile:
1.From setup , enter profiles in Quick Find box
2.Select profiles(Standard user).
3.Click clone.
4.For Profile, enter Sales Department.
5.Click save.


Profiles | Salesforce
orgfarm-0824836fa1-dev-ed.develop.lightning.force.com/lightning/setup/EnhancedProfiles/home
Developer Edition
Search Setup
Setup
Home
Object Manager
profile
Users
Profiles
Didn't find what you're looking for? Try using Global Search.
SETUP
Profiles
Profiles
Help for this Page
All Profiles
Edit | Delete | Create New View
New Profile
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Other All
Action
Profile Name
User License
Custom
Edit | Clone
Analytics Cloud Integration User
Analytics Cloud Integration User
Analytics Cloud Security User
Analytics Cloud Integration User
Anypoint Integration
Identity
Authenticated Website
Authenticated Website
Authenticated Website
Authenticated Website
B2B Reordering Portal Buyer Profile
External Apps Login
Chatter External User
Chatter External
Chatter Free User
Chatter Free
Chatter Moderator User
Chatter Free
Contract Manager
Salesforce
Cross Org Data Proxy User
XOrg Proxy User
Custom: Marketing Profile
Salesforce
1-25 of 45
0 Selected
Previous
Next
Page 1 of 2
Hot weather
Now
Search
ENG
IN
12:15
06-03-2026

Activity 2:

Create a custom profile- 2:
Create a profile with the profile name as " Farm Manager" .
6. From setup , enter profiles in Quick Find box
7.Select profiles (Standard user).
8.Click clone.

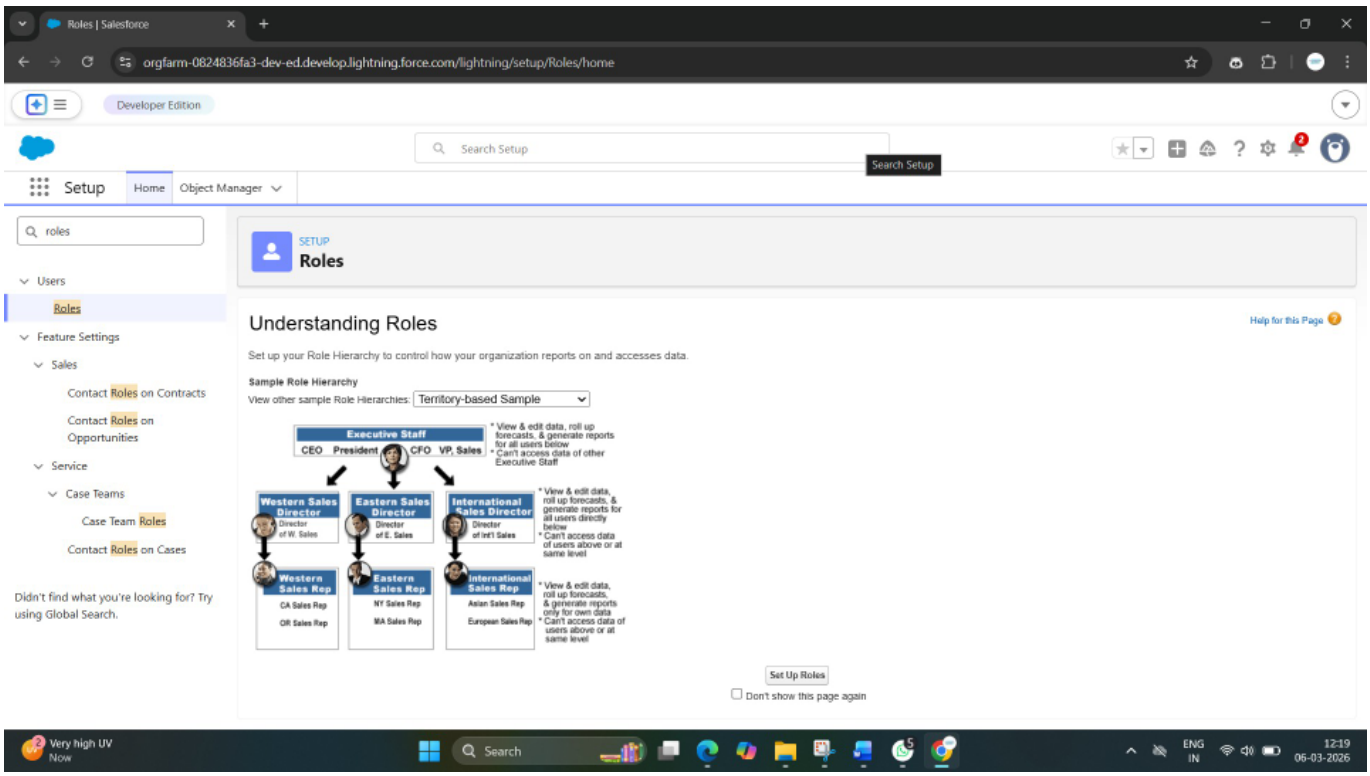
Milestone 7- Role

In Salesforce, roles are used to determine which users have access to certain data and functions within the system. They are also used to define the reporting hierarchy within an organization. Users with higher roles have greater access to data and more control over the system.

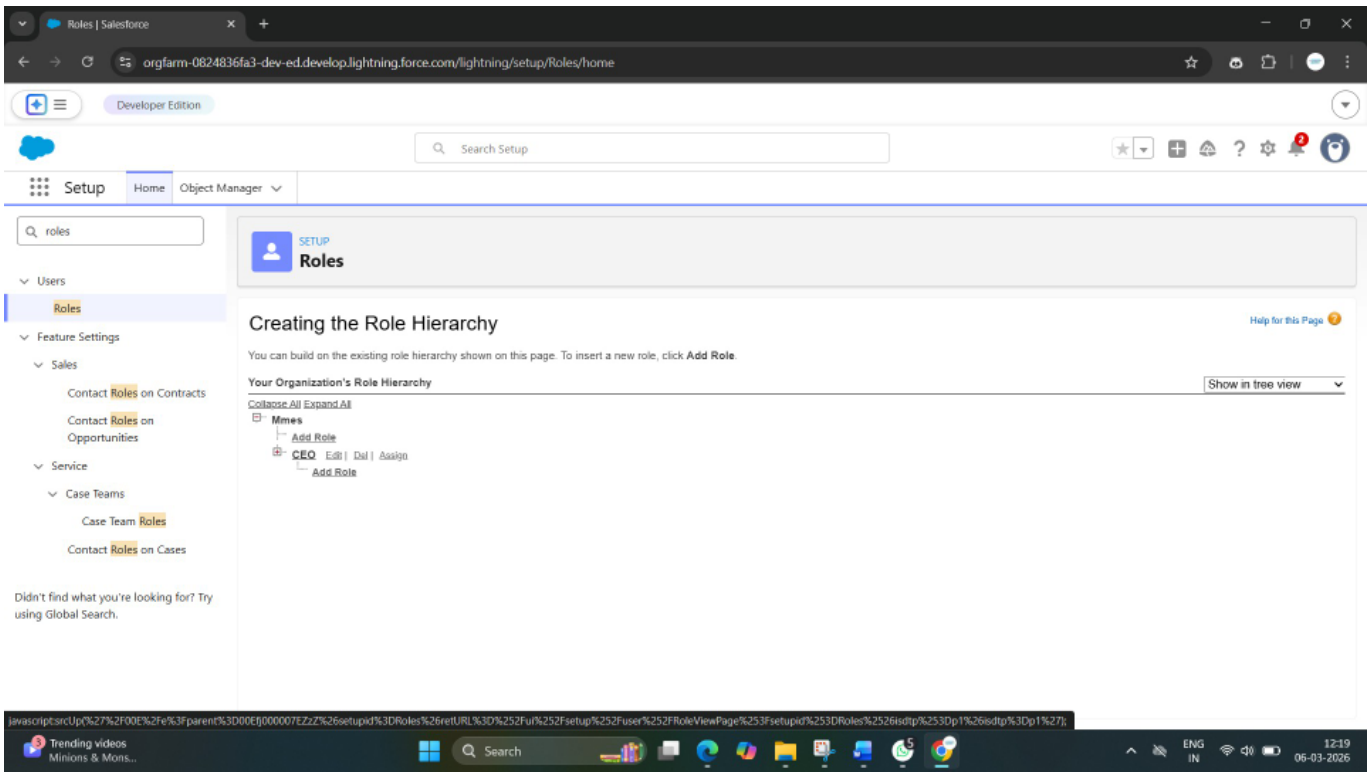
Activity- 1

Creation of role

1. From the Quick find box search for the role and click on the roles option 2. select the set up roles option



3.Below the CEO click on add role and enter the label name as a " Farm Manager " and role name will be Automatically populated and click on save.



Milestone 8- User :

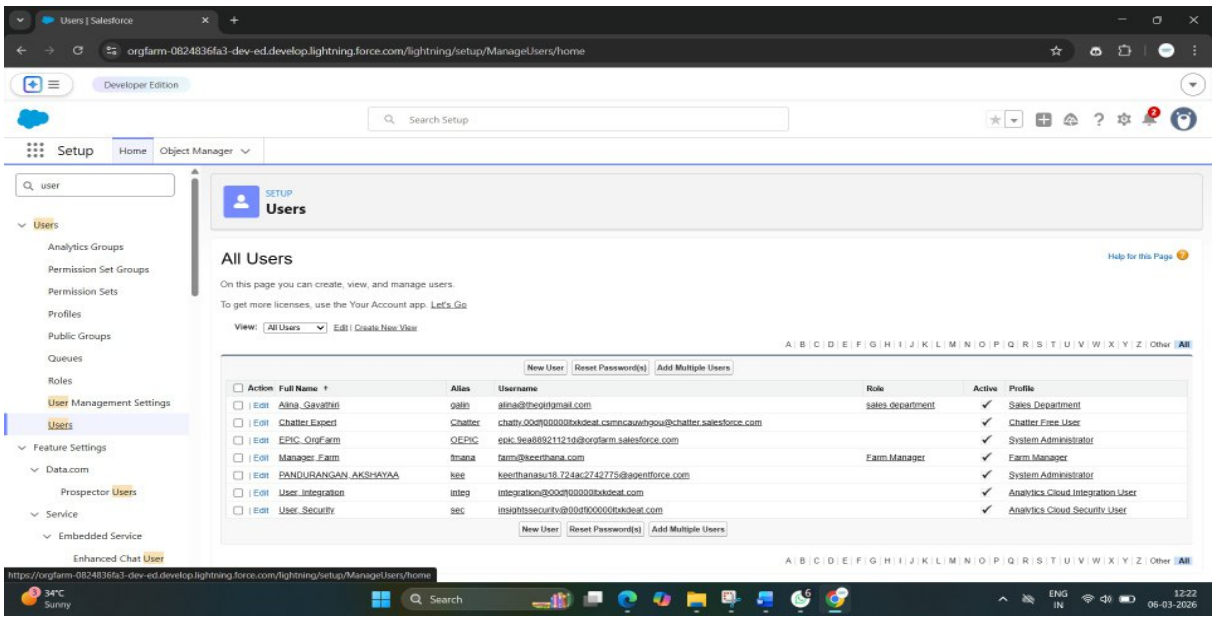
What is a user?
A user is anyone who logs into Salesforce. Users are employees at your company, such as sales reps, managers, and IT specialists, who need access to the company's records. Every user in Salesforce has a user account. The user account identifies the user, and the user account settings determine what features and records the user can access.

NOTE- As Salesforce license can only be used by 2 Users at a time in Dev Org, so If you don't find salesforce license then deactivate a user who has salesforce license Or change the license type from Salesforce to any other.

Activity 1:

To Create a user:

1. From Setup, enter Users in the Quick Find box, then select Users.
2. Click New User.



3.Enter First name as Farm and last name as Manager.
4.Enter the user's name and email address and a unique username in the form of an email address. By default, the username is the same as the email address.
5.Select the role Farm Manager.
6.Select user License as Standard Platform User.
7.Select profile(Farm Manager).
8.Click save.

Activity 2:
Create a user with a username as "Sales Department", and assign him the sales Manager profile. Follow the steps from above Activity

Milestone 9- Sharing Rules:

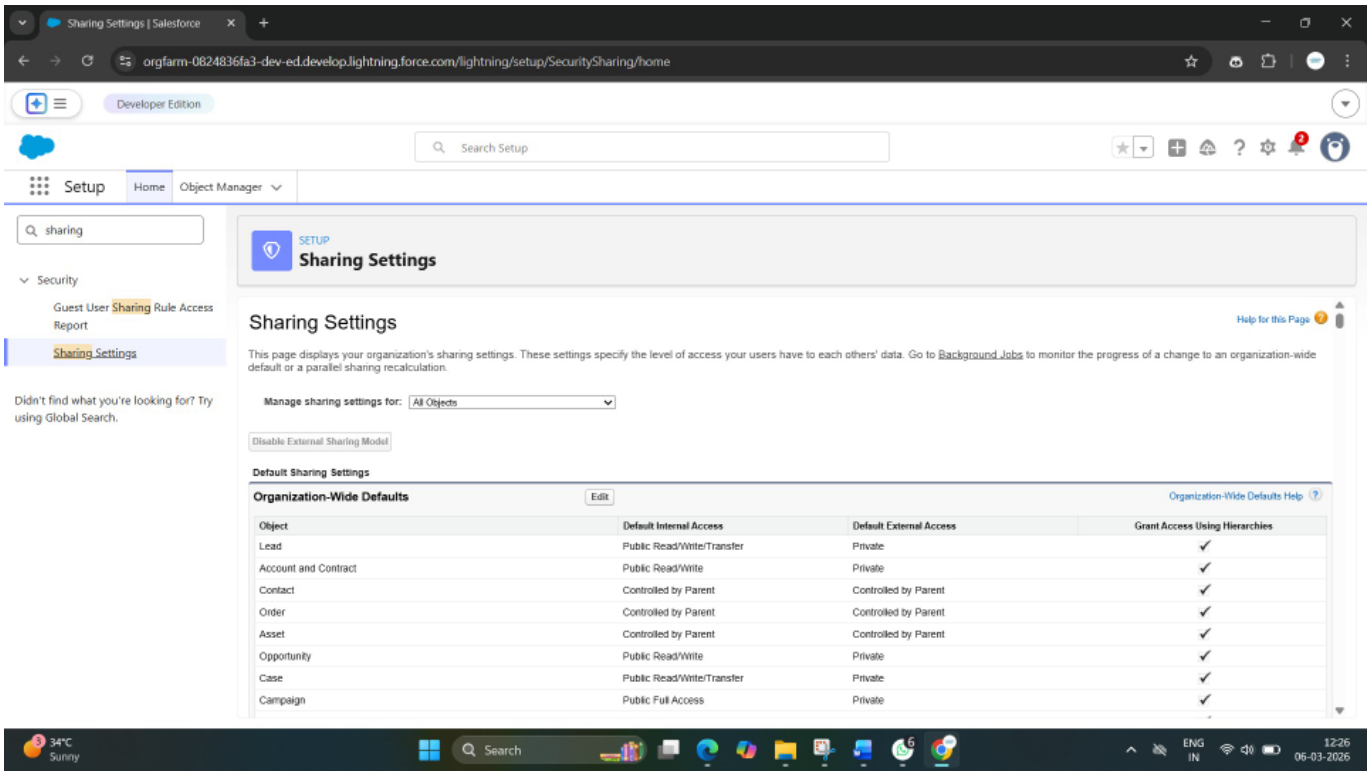
What are Sharing Rules?
Sharing rules help users to share records based on conditions. It is basically created for objects whose organization- wide defaults (OWD) are set to public read- only or private because sharing rules can only extend the access and not restrict it.

Types of sharing rules,
•Owner- based Sharing Rules
•Criteria- based Sharing Rules

Activity 1:

Create a sharing rule
1)Goto Sharing Settings, which can be found under the Quick Find section.
2)Scroll down and find the candidate object where a sharing rule needs to be added, and then click on New to create a new sharing rule.
3)Add the label of the sharing rule you want to make.
4)Select your rule type based on the criteria.
5)Select the field can join immediately check field from the Crop object.

6) Select the Crop as equal and value is True.
7) And in selecting the users to share with the section select roles and in that select Farm Manager.
8) And in the section of select the level of access for the users give the access Read/ Write.
9) And save the rule.



Activity 2:
Create a Sharing Rule to Share the records of Buyer to Sales department with the Access of Read/ Write.
Follow the steps from above Activity.

Milestone- 10: User Adoption

Activity 1:

Create a Record (Farmer)

- Click on App Launcher on left side of screen.
- Search Farm Management & click on it.
- Click on Farmer Tab.
- Click new and fill details & Save.

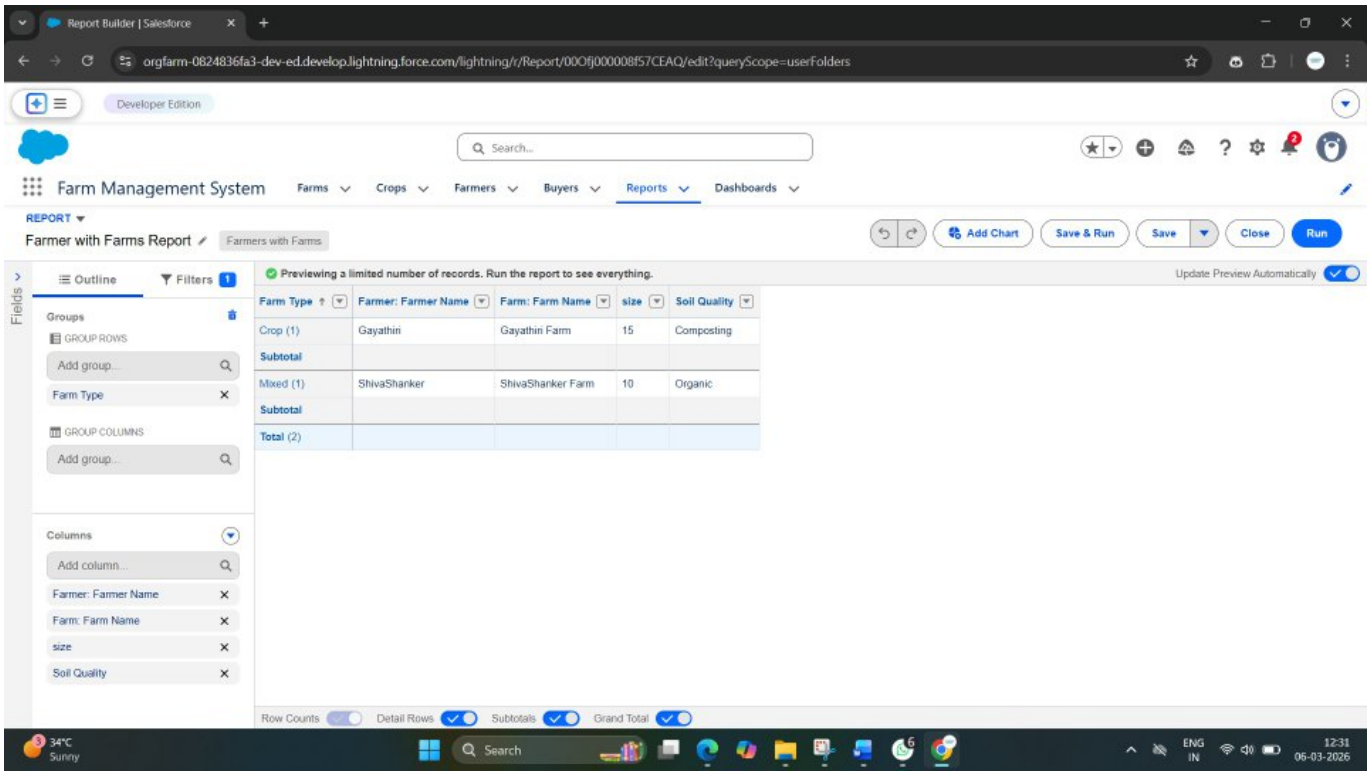
Milestone 11- Reports:

What are Reports?
A report is a list of records that meet the criteria you define. It's displayed in rows and columns, and can be filtered, grouped, or displayed in a graphical chart. Every report is stored in a folder. Folders can be public, hidden, or shared, and can be set to read-only or read/write.

Activity 1:

Create a report:
Create a report that displays rating of the account and which has type and account name.

1) Click on app launcher search for reports.
2) Click on the new report and select the category has Farmer with Farms.
3) In the details section select the option start report.
4) In Outline Click on Column Select Which Field you want to display in the Report.
5) Click Group, Select Which Fields You Want to Group.
6) Save the report by giving label name(Farmer with Farms) and save the folder as a public folder and save the report.



Activity 2:
Create a Report using the Objects Crops, Buyers. Follow the steps from above Activity.

Milestone 12- Dashboard:

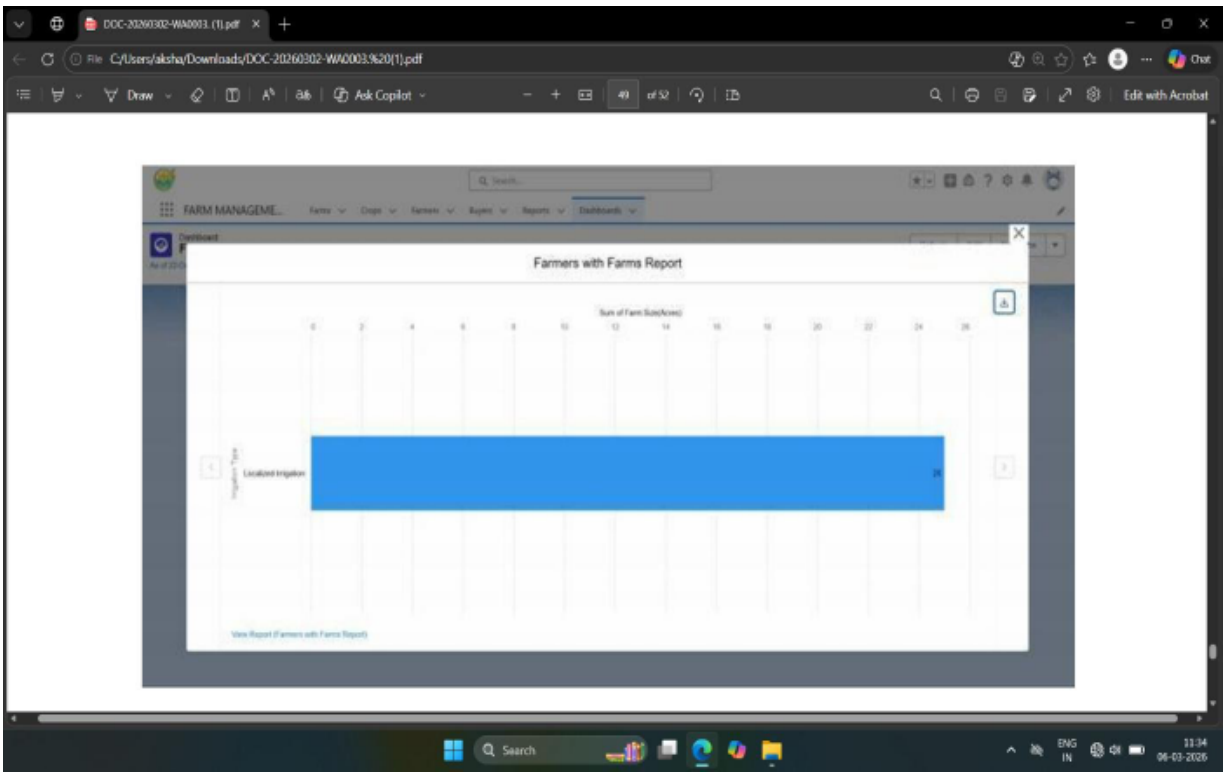
Dashboards provide more insights than reports as they combine the data from many

reports and show a summarized result. Looking at many reports at a time gives the flexibility of combining the results from them quickly. Also summaries in dashboards help us decide on action plans quicker. The dashboards can contain charts, graphs and Tabular data.

Activity 1:

Create a Dashboard:

1. Click the Dashboard tab.
2. Click New Dashboard.
3. Name the dashboard Farmers with farms and click Create.
4. Click + Component.
5. Select the Farmers with Farms and click Select.
6. Select the Gauge chart and click Add.
7. Click Save and then Done.

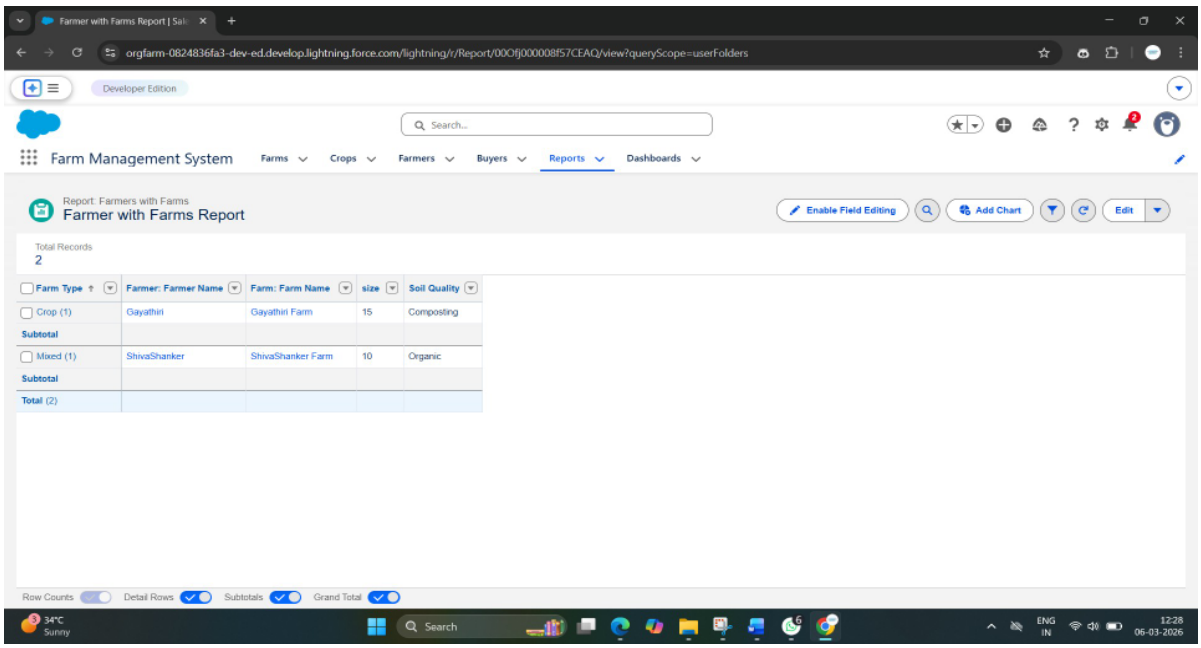


Milestone- 13: View Report and Dashboard:

Activity 1:

Report

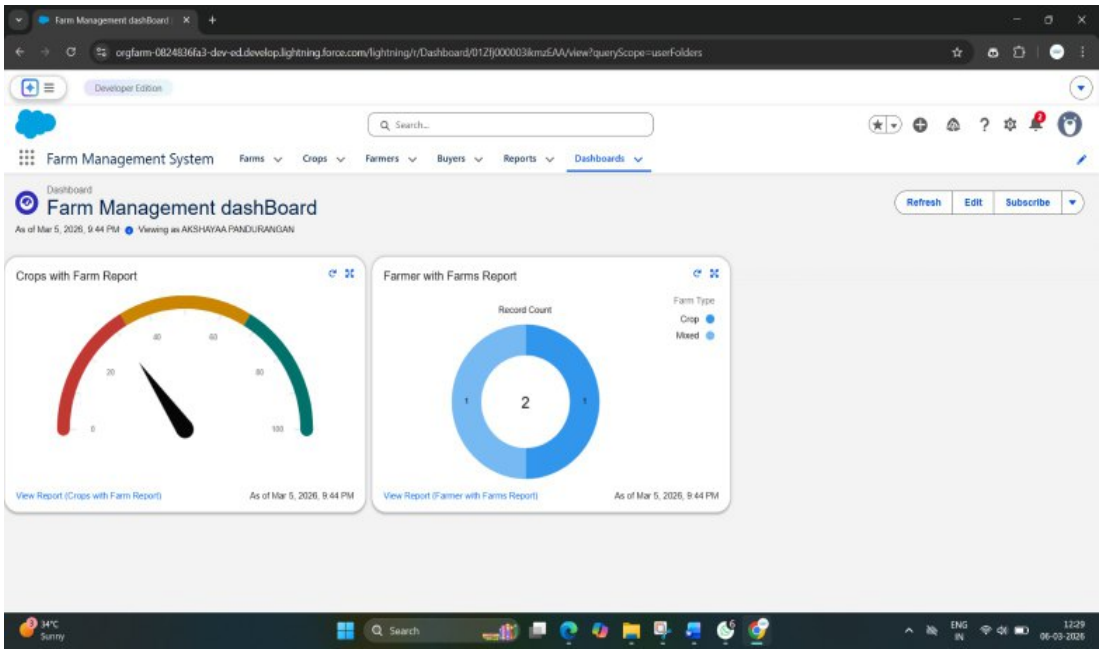
- Click on App Launcher on left side of screen.
- Search Farm Management System & click on it.
- Click on Reports Tab.
- Click on Farmers with Farms & see records



Activity 2:

Dashboard

•Click on App Launcher on left side of screen.
•Search Farm Management system & click on it.
•Click on Dashboard Tab.
•Click Farmers With farms & see records.



8. FUNCTIONAL & PERFORMANCE TESTING

During the testing phase, screenshots were captured from the Salesforce interface to

verify system functionality. The following were tested:

- Creation of Farm, Crop, Farmer, and Buyer records
- Validation of input fields
- Relationship and dependency checks
- Profile and role access testing
- Sharing rule verification
- Report and dashboard generation

Performance testing ensured smooth system response and stability during multiple user operations.

8.1 ADVANTAGES & DISADVANTAGES

Advantages:

Reduced paperwork and manual records

- Improved data accuracy
- Better monitoring of farm activities
- Real-time analytics and reports
- Easy access to farm information
- Better decision making for farmers
- Organized management of crops and buyers
- Faster data processing
- Secure cloud-based storage using Salesforce

Disadvantages:

- Requires internet connection to access the system
- Dependency on the Salesforce platform
- Limited offline access
- Requires basic technical knowledge
- Initial setup and configuration time
- Data security concerns if not managed properly
- System maintenance may be required
- Dependence on cloud services
- Training needed for new users
- Possible cost for advanced features or licenses

9.Conclusion:

The Farm Management System (FMS) developed using Salesforce successfully provides a digital

platform for managing agricultural data and farm operations. The system helps farmers and farm managers organize important information such as farm details, crop records, farmer profiles, and buyer information in a structured and secure manner.
By using Salesforce features like objects, reports, dashboards, and automation tools, the system improves the efficiency of farm management activities. It reduces manual paperwork, minimizes human errors, and ensures that data can be easily accessed and updated whenever required.
The Farm Management System also supports better decision- making by providing useful insights through reports and dashboards. Farm managers can monitor crop performance, track farm activities, and manage buyer relationships more effectively. Overall, the system helps in modernizing agricultural management by bringing technology into traditional farming practices.

10.Future Scope:

Mobile application integration
- AI - based crop prediction
- Weather data integration
- IoT sensor connectivity
- Government scheme integration
- Market price forecasting
- Smart irrigation management
- Automated farm alerts and notifications
- Integration with digital payment systems
- Advanced data analytics for crop productivity

11. Appendix:

- Source Code: Salesforce Configuration and Metadata
- Dataset: Farm, Crop, Farmer, and Buyer Records
- Tools Used: Salesforce Developer Edition
- Reports and Dashboards Configuration
- Custom Objects: Farm, Crop, Farmer, Buyer
- Automation Tools: Workflow Rules and Process Builder
- Test Data and Sample Records

THANK YOU